



HCMUS
Viet Nam National University
Ho Chi Minh City
University of Science

CSC10006 – Introduction to Database

LAB TUTORIALs

Installation guide for MSSQL

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1 Objectives

- Installation and hands-on work with SQL Server Management Studio (SSMS), including: how to install the software; how to create databases and enter data; how to generate database creation scripts; how to import and export data; and how to attach and detach databases.
- After completing this exercise, students will be able to:
 - o Install Microsoft SQL Server
 - o Start and stop the required SQL Server services
 - o Perform basic operations such as creating databases, creating tables, defining primary keys and foreign keys, and entering data
 - o Attach and detach databases
 - o Generate scripts for database creation
 - o Import and export databases to and from other databases
- Link download:
 - o <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>
 - o SQL Management Studio:
https://go.microsoft.com/fwlink/?linkid=87312_6

2 Detailed Instructions

2.1 Install SQL server instance

There are 2 ways to install SQL Server instance:

- **SQL Server for Windows:** Only applicable for Windows users
- **SQL Server for Linux:** Applicable for all platforms such as Linux, Mac, Windows via Docker

You can freely choose any way to use SQL Server in almost lab assignments with an exception that some assignments can only be done on Windows (SQL Management Studio), please complete them in lab. It is recommended that you should try both ways.

2.2 SQL Server for Windows (2022)

In this tutorial, you will learn to install the SQL Server 2022 Developer Edition and SQL Server Management Studio (SSMS)

2.2.1 Install SQL Server 2022 Developer Edition

- Download Sql Server 2022 Developer Edition



Developer

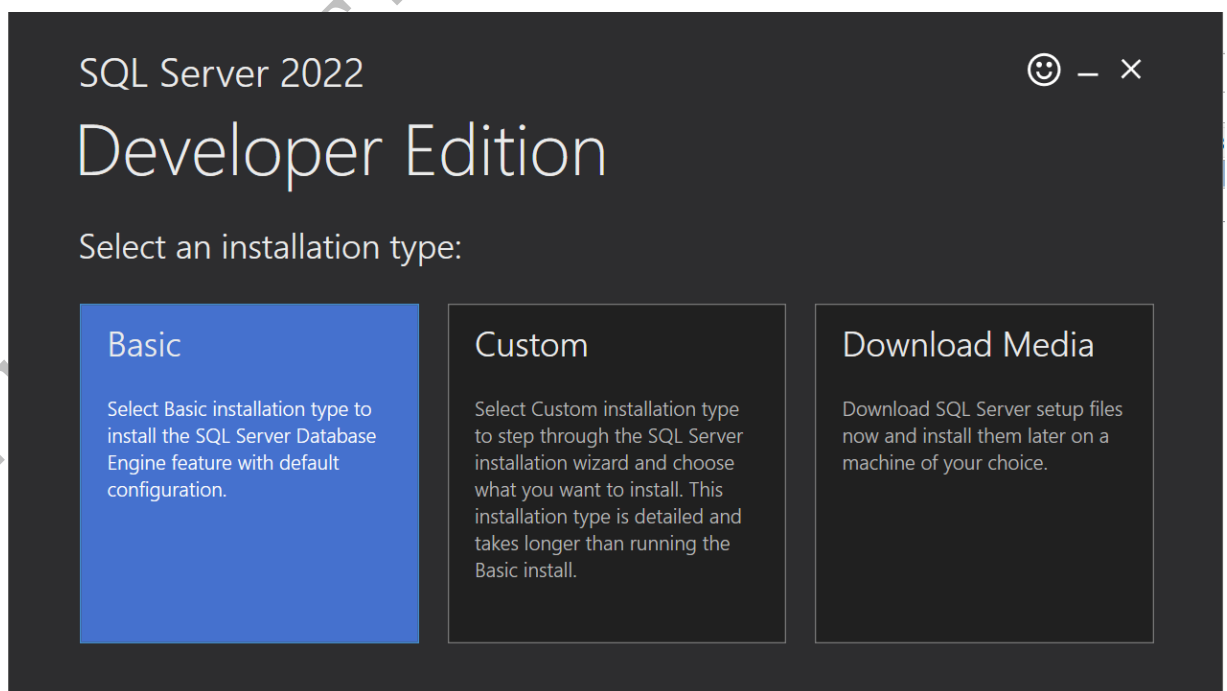
SQL Server 2022 Developer is a full-featured free edition, licensed for use as a development and test database in a non-production environment.

[Download now](#)

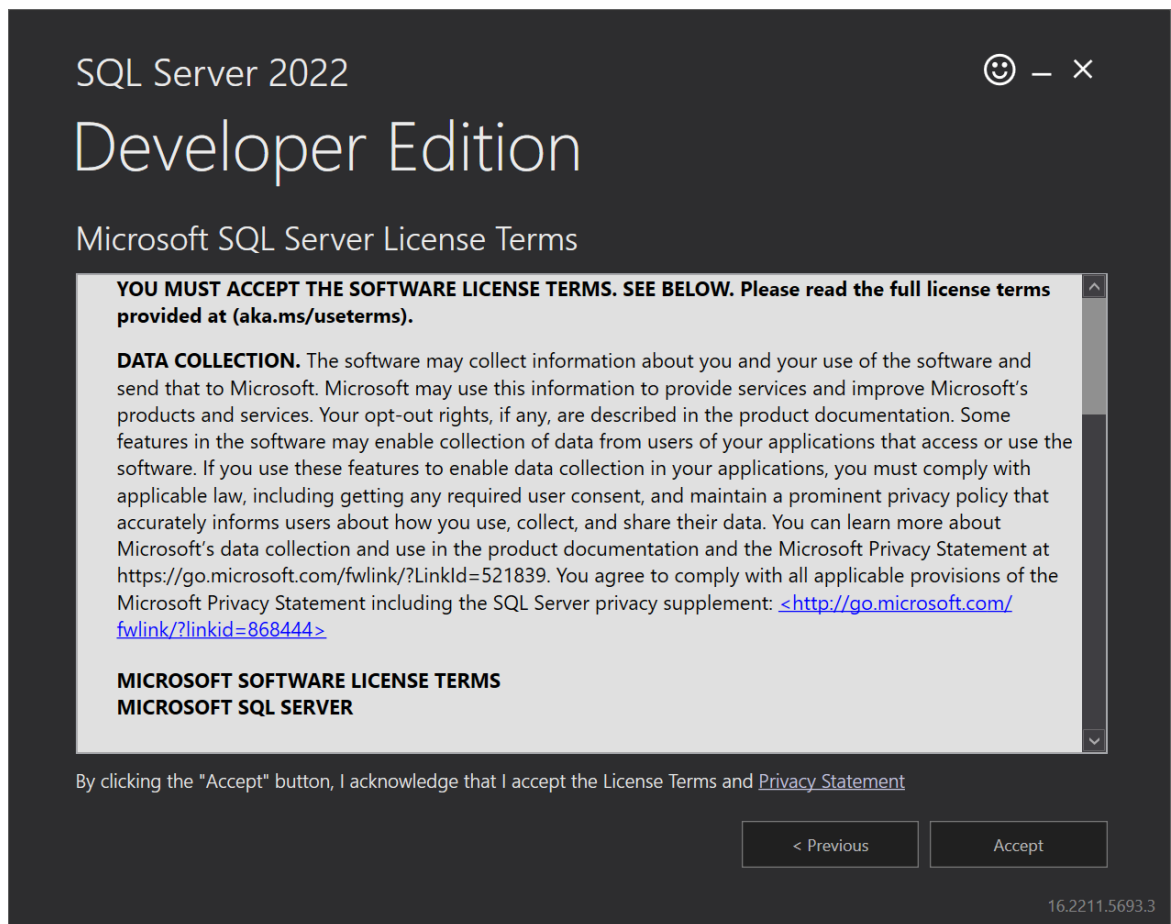
- Double-click the file SQL2022-SSEI-Dev.exe file to launch the downloader.

▼ Today (1)

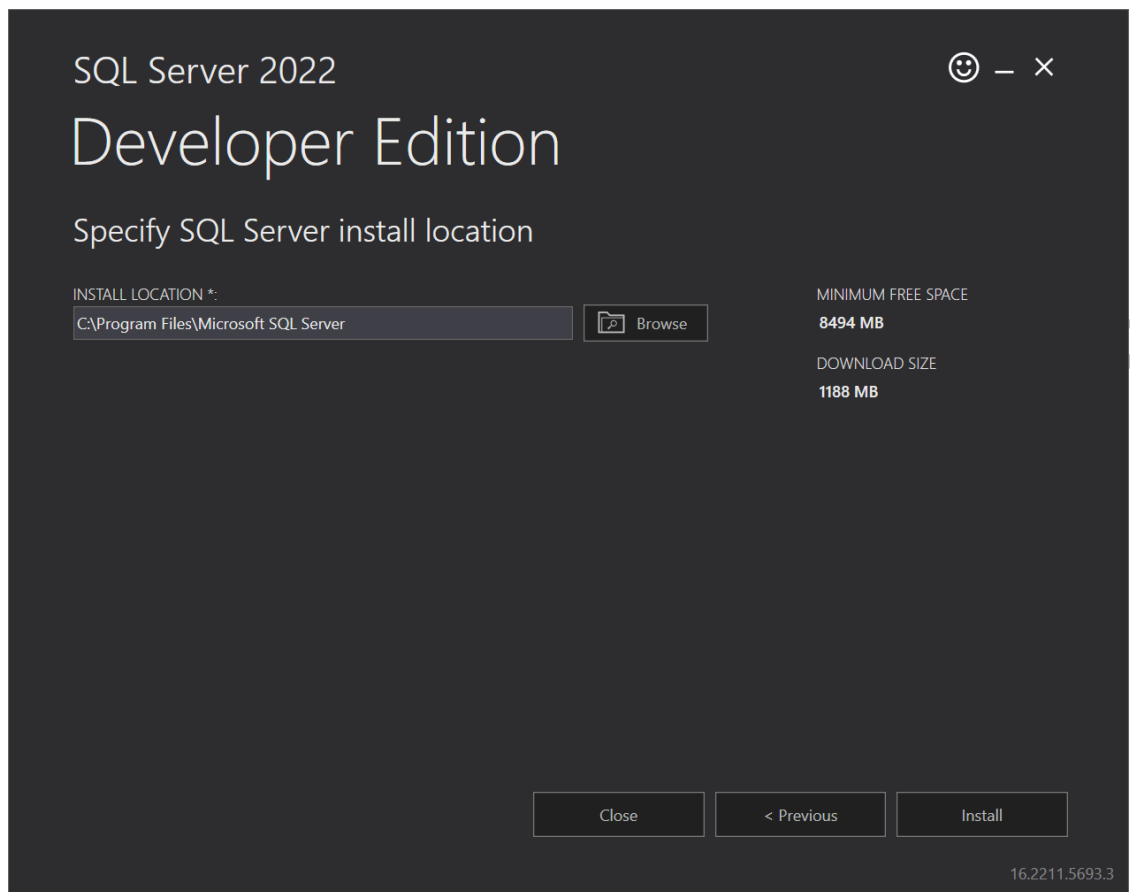
 SQL2022-SSEI-Dev



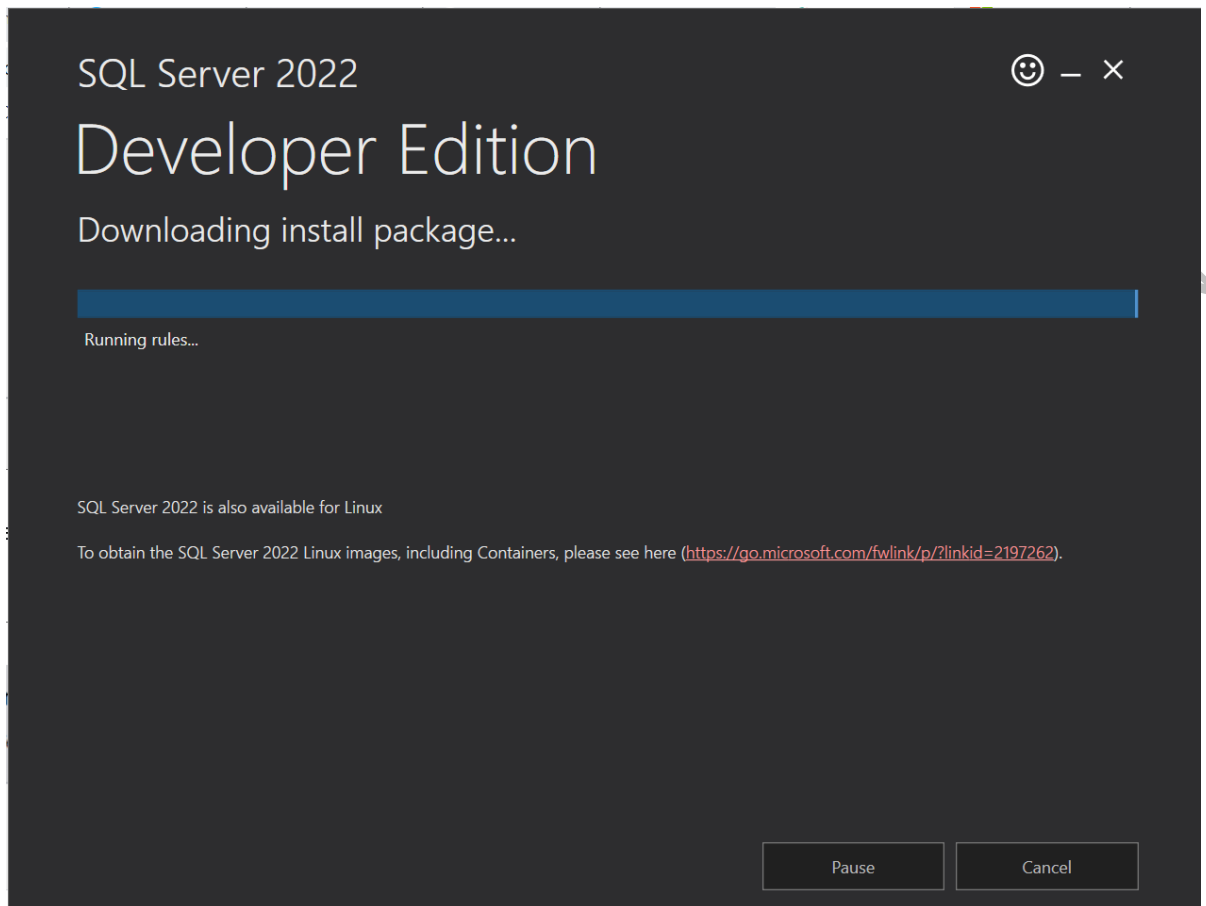
- The downloader will ask you to select the installation type, choose **Basic**.



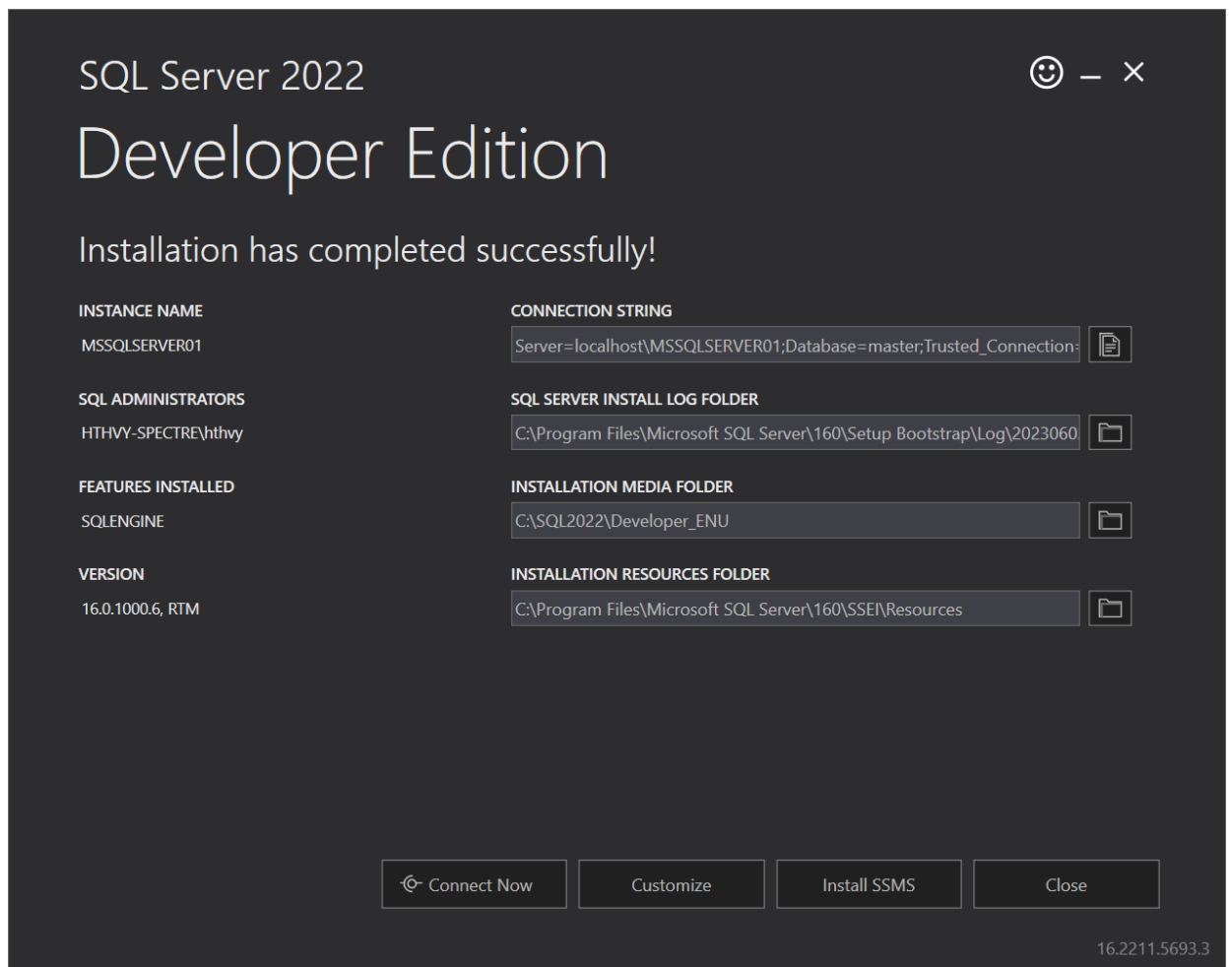
- Click **Accept** button > Specify SQL server install location.



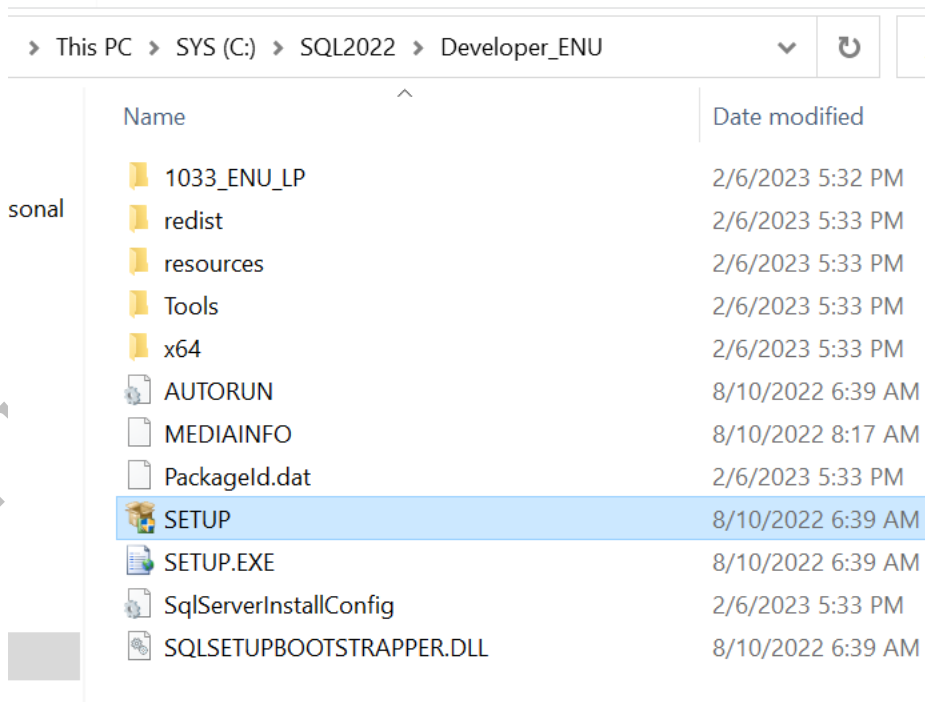
- Then click **Install** button.



- The downloader will start downloading the installation files. It could take a while.



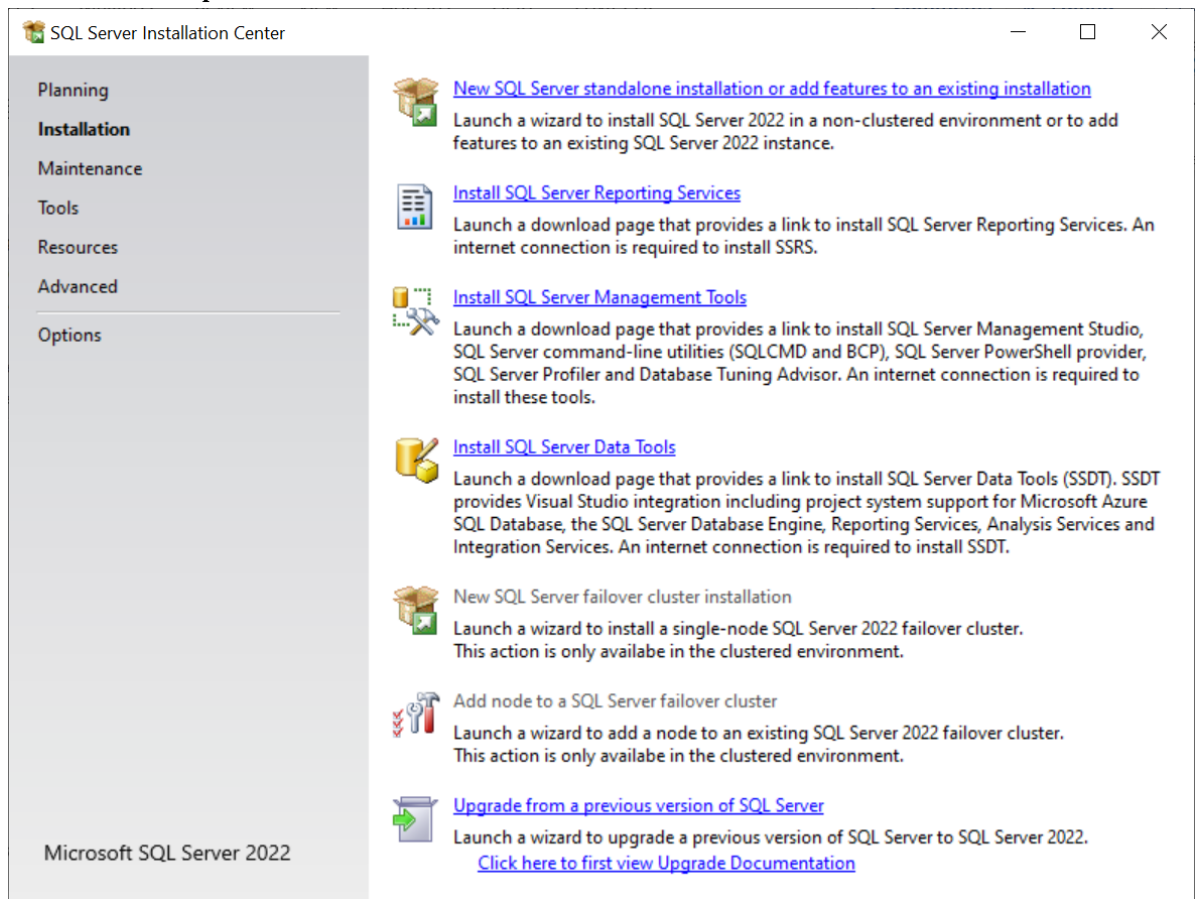
- Once the download completes, open the folder that stores the downloaded file.



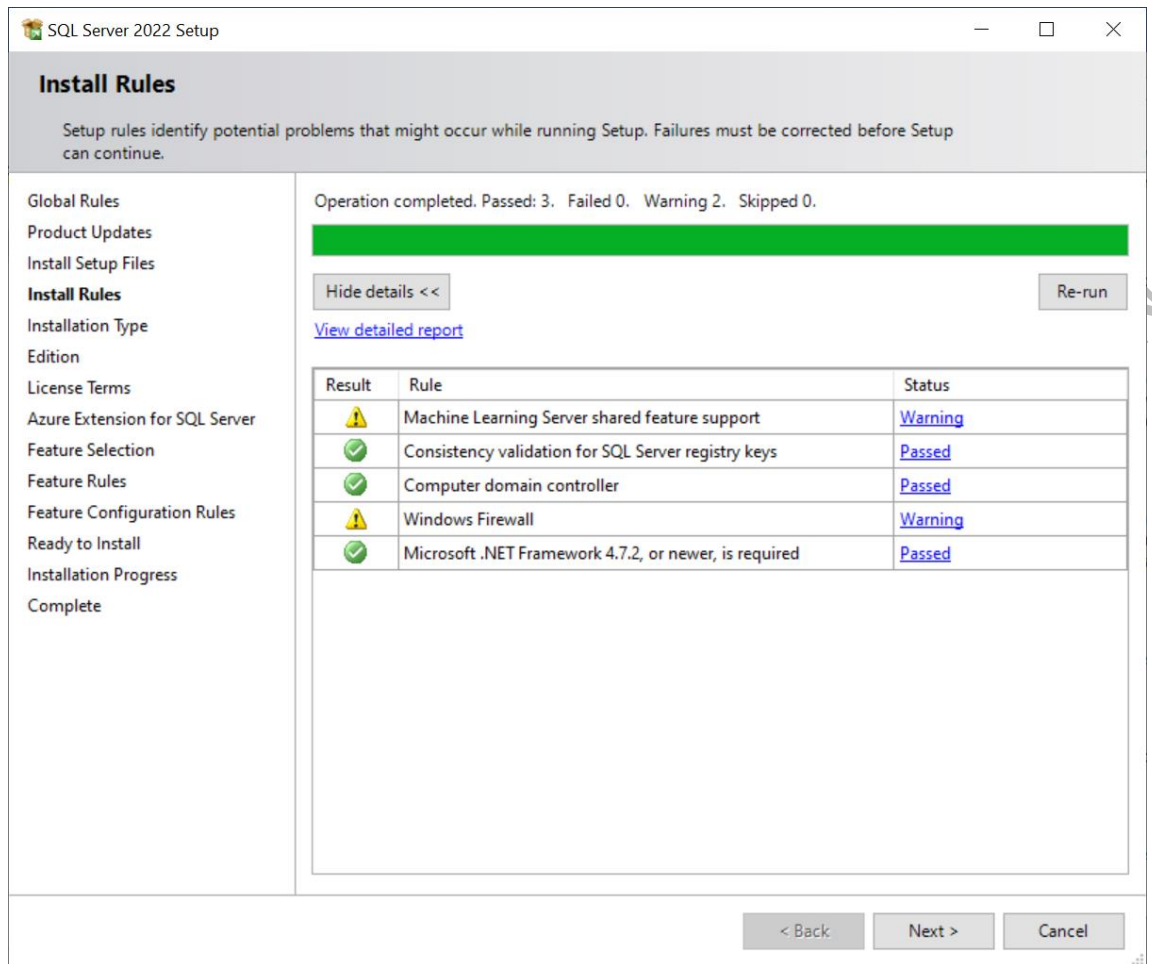
- Open the installation folder and click the **setup.exe** file to launch the installer

- **Install SQL Server 2022 developer edition**

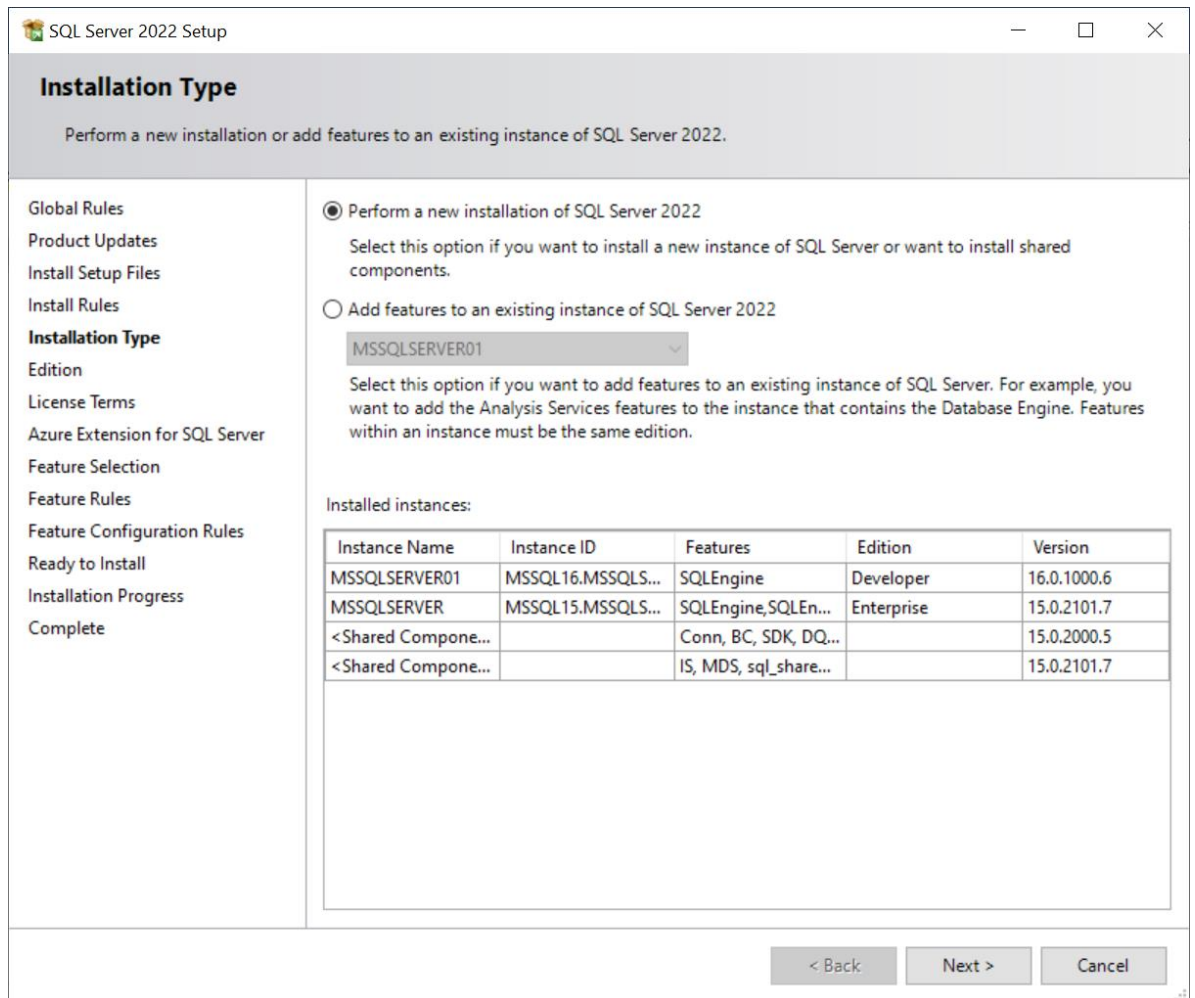
- After double click setup.exe, you will see the following window: select the installation option on the left:



- Click the first link to launch a wizard to install SQL Server 2022 > **Next**.



- Click Next > New installation



- Specify the edition that you want to install, select **Developer** edition, and click the **Next** button

SQL Server 2022 Setup

Edition

Select the edition of SQL Server 2022 you want to install.

- Global Rules
- Product Updates
- Install Setup Files
- Install Rules
- Installation Type
- Edition**
- License Terms
- Azure Extension for SQL Server
- Feature Selection
- Feature Rules
- Feature Configuration Rules
- Ready to Install
- Installation Progress
- Complete

Select an edition of SQL Server to install. You can choose to either use a SQL Server license that you have already purchased by entering the product key or choose pay-as-you-go billing through Microsoft Azure. You can also specify a free edition of SQL Server: Developer, Evaluation, or Express. Evaluation has the largest set of SQL Server features, as documented in SQL Server Books Online, and is activated with a 180-day expiration. Developer edition does not have an expiration, has the same set of features found in Evaluation, but is licensed for non-production database application development only. To upgrade from one installed edition to another, run the Edition Upgrade Wizard.

☒ Specify a free edition:

Developer

☐ Use pay-as-you-go billing through Microsoft Azure:

Warning: To enable this option, you must have an active Azure subscription that you will be required to provide along with a resource group, Azure region, and tenant ID later in setup. For more information, see <https://aka.ms/ArcEnabledSqlPAYG>.

Standard

☐ Enter the product key:

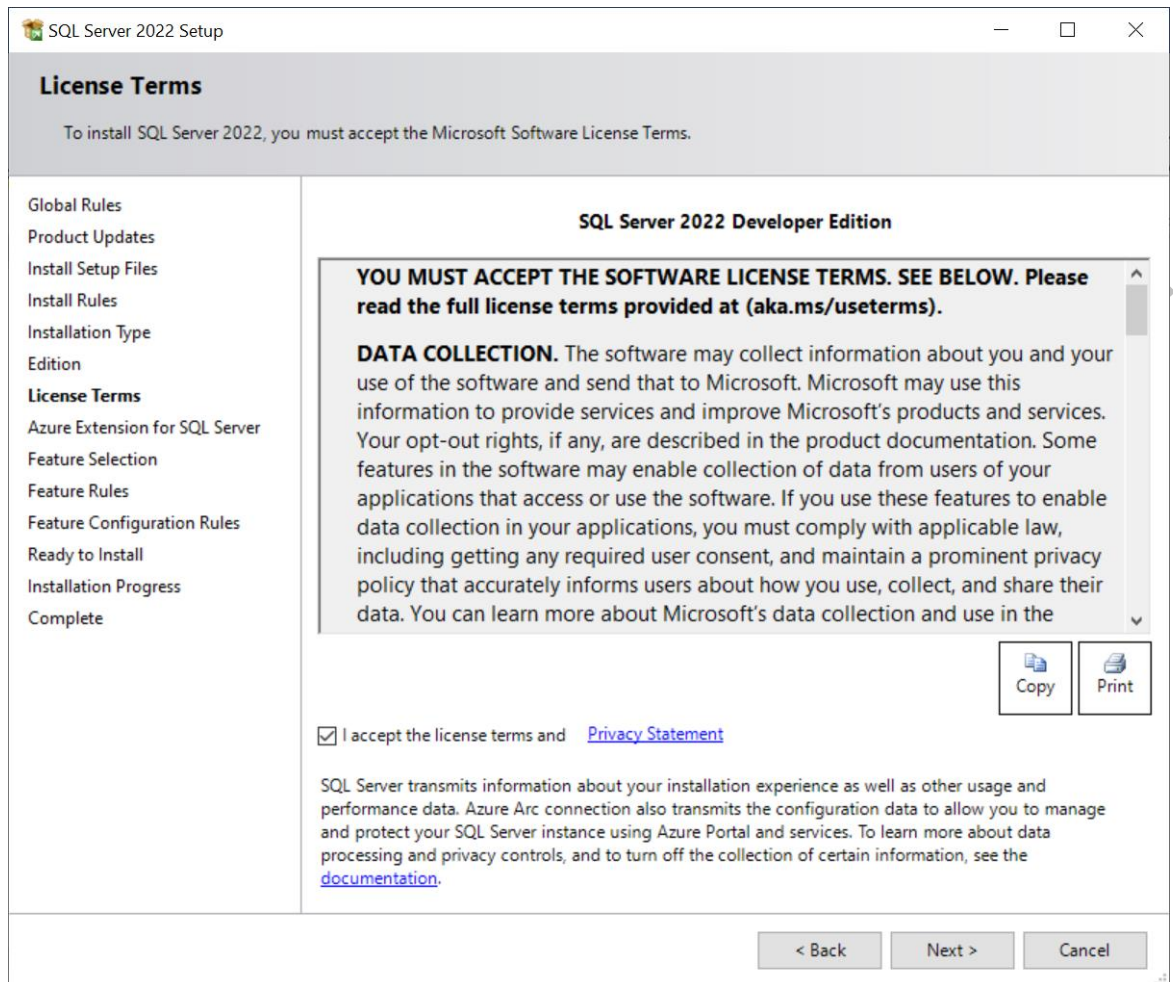
- - - -

☐ I have a SQL Server license with Software Assurance or SQL Software Subscription

☐ I have a SQL Server license only

< Back Next > Cancel

- Select the “I accept the license terms.” and click the Next button:



- Uncheck the “Azure extension for SQL server” if you do not want and click the Next button

SQL Server 2022 Setup

Azure Extension for SQL Server

Azure Extension for SQL Server is required to enable Microsoft Defender for Cloud, Purview, and Azure Active Directory.

Global Rules
Product Updates
Install Setup Files
Install Rules
Installation Type
Edition
License Terms
Azure Extension for SQL Serv...
Feature Selection
Feature Rules
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

☐ Azure Extension for SQL Server

To install Azure extension for SQL Server, provide your Azure account or a service principal to authenticate the SQL Server instance to Azure. You also need to provide the Subscription ID, Resource Group, Region, and Tenant ID where this instance will be registered. For more information for each parameter, visit <https://aka.ms/arc-sql-server>.

☐ Use Azure Login
☒ Use Service Principal

Azure Service Principal ID*

Azure Service Principal Secret*

Azure Subscription ID*

Azure Resource Group*

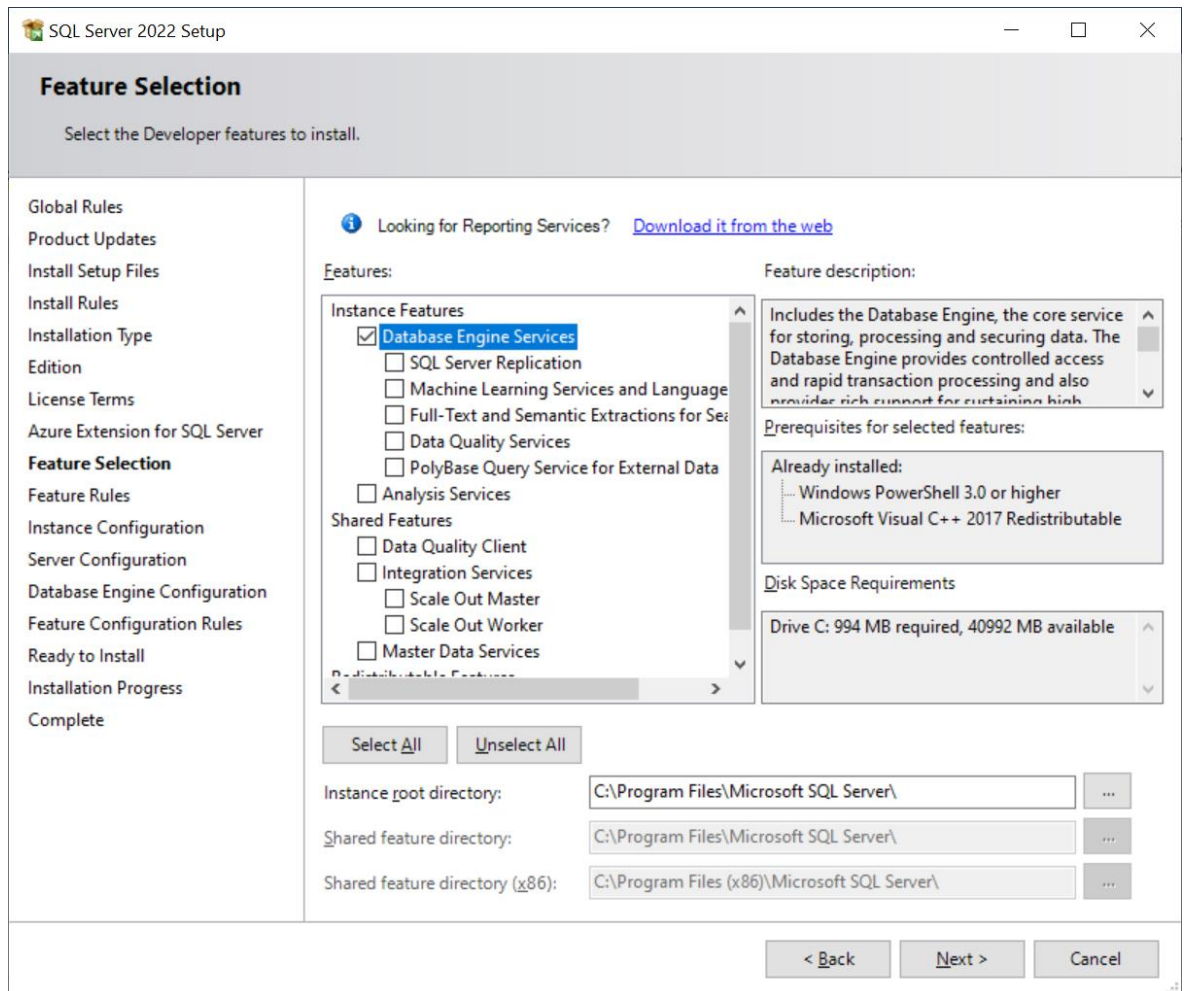
Azure Region*

Azure Tenant ID*

Proxy Server URL (optional)

< Back
Next >
Cancel

- Select the features that you want to install. For learning purposes, you need the **Database Engine Services**; check the checkbox and click the Next button to continue



- Provide the instance ID of the SQL Server and click the Next button

SQL Server 2022 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Global Rules

Product Updates

Install Setup Files

Install Rules

Installation Type

Edition

License Terms

Azure Extension for SQL Server

Feature Selection

Feature Rules

Instance Configuration

Server Configuration

Database Engine Configuration

Feature Configuration Rules

Ready to Install

Installation Progress

Complete

☐ Default instance

☒ Named instance: *

Instance ID:

SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQL2022

Installed instances:

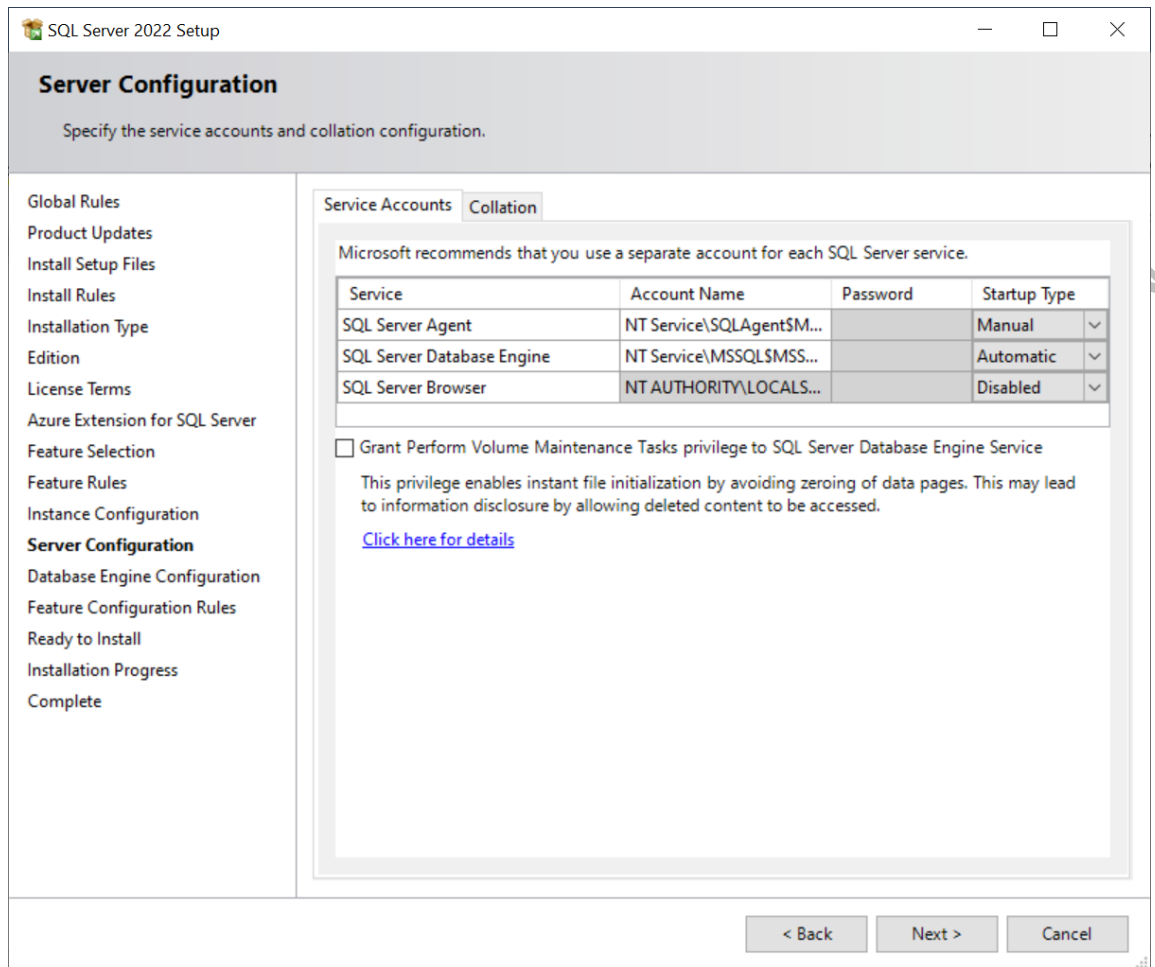
Instance Name	Instance ID	Features	Edition	Version
MSSQLSERVER01	MSSQL16.MSSQLS...	SQLEngine	Developer	16.0.1000.6
MSSQLSERVER	MSSQL15.MSSQLS...	SQLEngine, SQLEn...	Enterprise	15.0.2101.7
<Shared Compone...		Conn, BC, SDK, DQ...		15.0.2000.5
<Shared Compone...		IS, MDS, sql_share...		15.0.2101.7

< Back

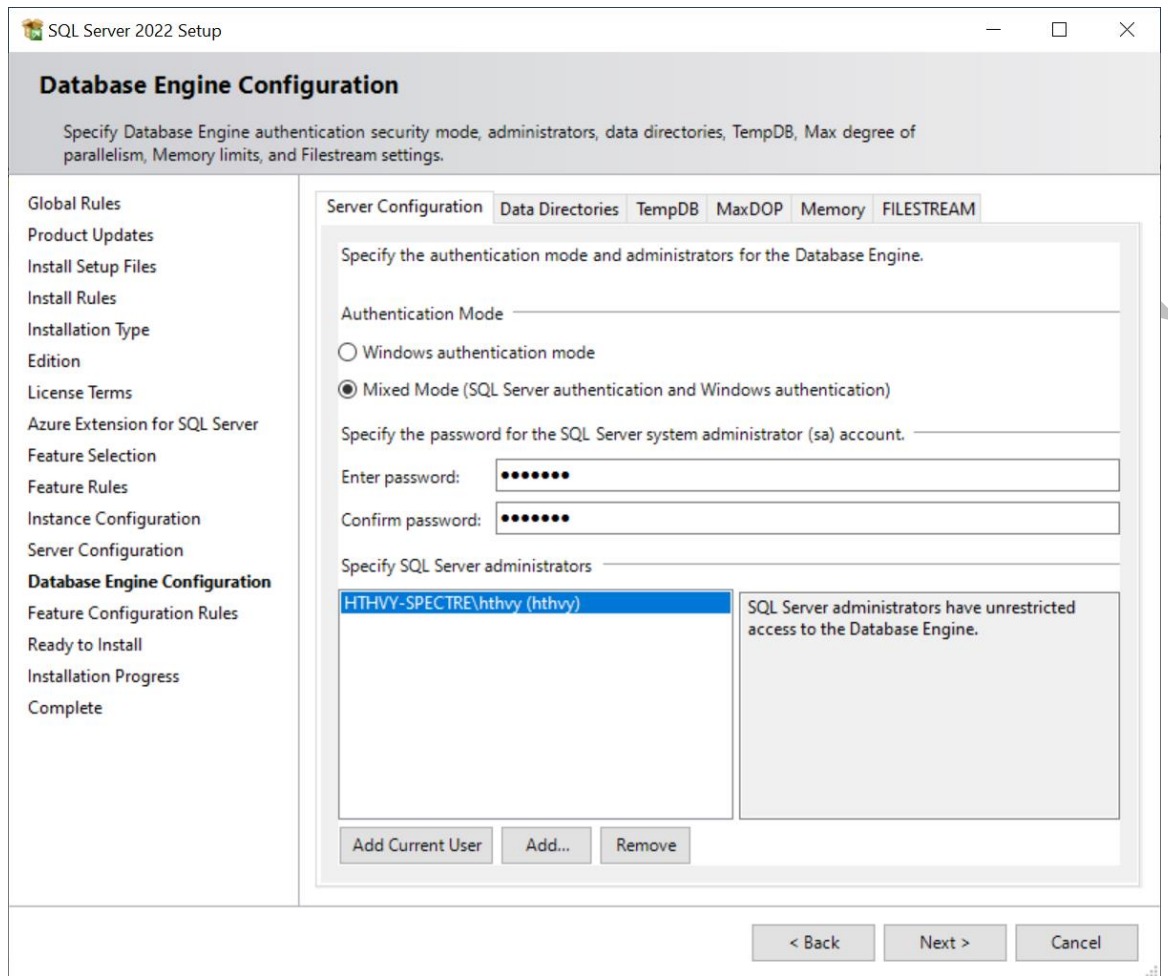
Next >

Cancel

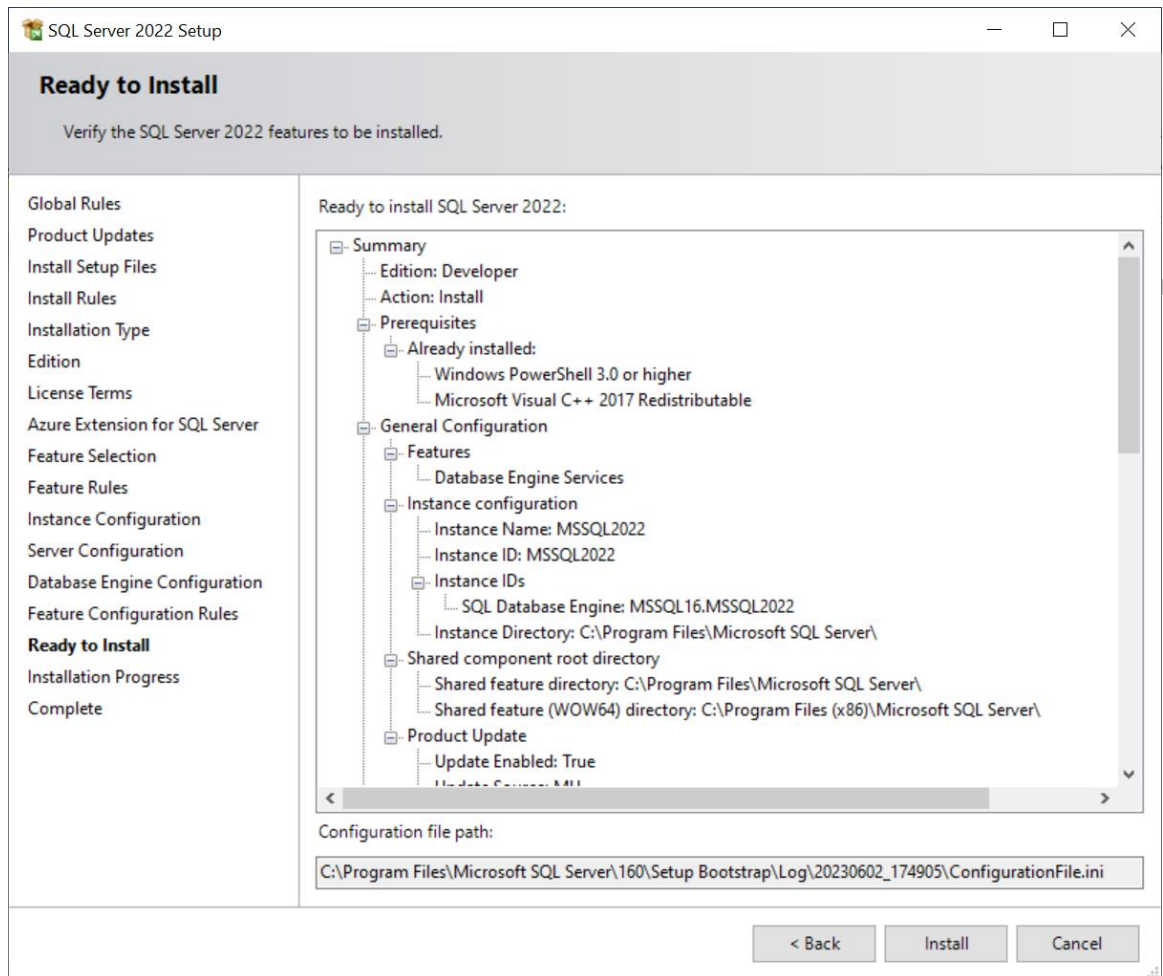
- Click Next button to continue



- Select the Mixed Mode, provide the password for system administration (sa) account (you need to store this password in a secure place so that you can use it to connect to the SQL Server later), click the Add Current User to specify the SQL Server Administrators, and click the Next button



- Verify the SQL Server 2022 features to be installed:



- Click the Close button to complete the installation

2.2.2 Install Microsoft SQL Server Management Studio

To interact with SQL Server, you need to have a SQL Server client tool. Microsoft provides you with the SQL Server Management Studio (SSMS). The SQL Server Management Studio is software for querying, designing, and managing SQL Server on your local computer or in the cloud. It provides you with tools to configure, monitor, and administer SQL Server instances.

- First, download the SSMS from the Microsoft website

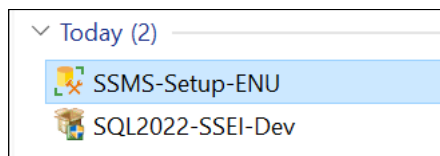
Download SSMS

↓ [Free Download for SQL Server Management Studio \(SSMS\) 19.1](#) ↗

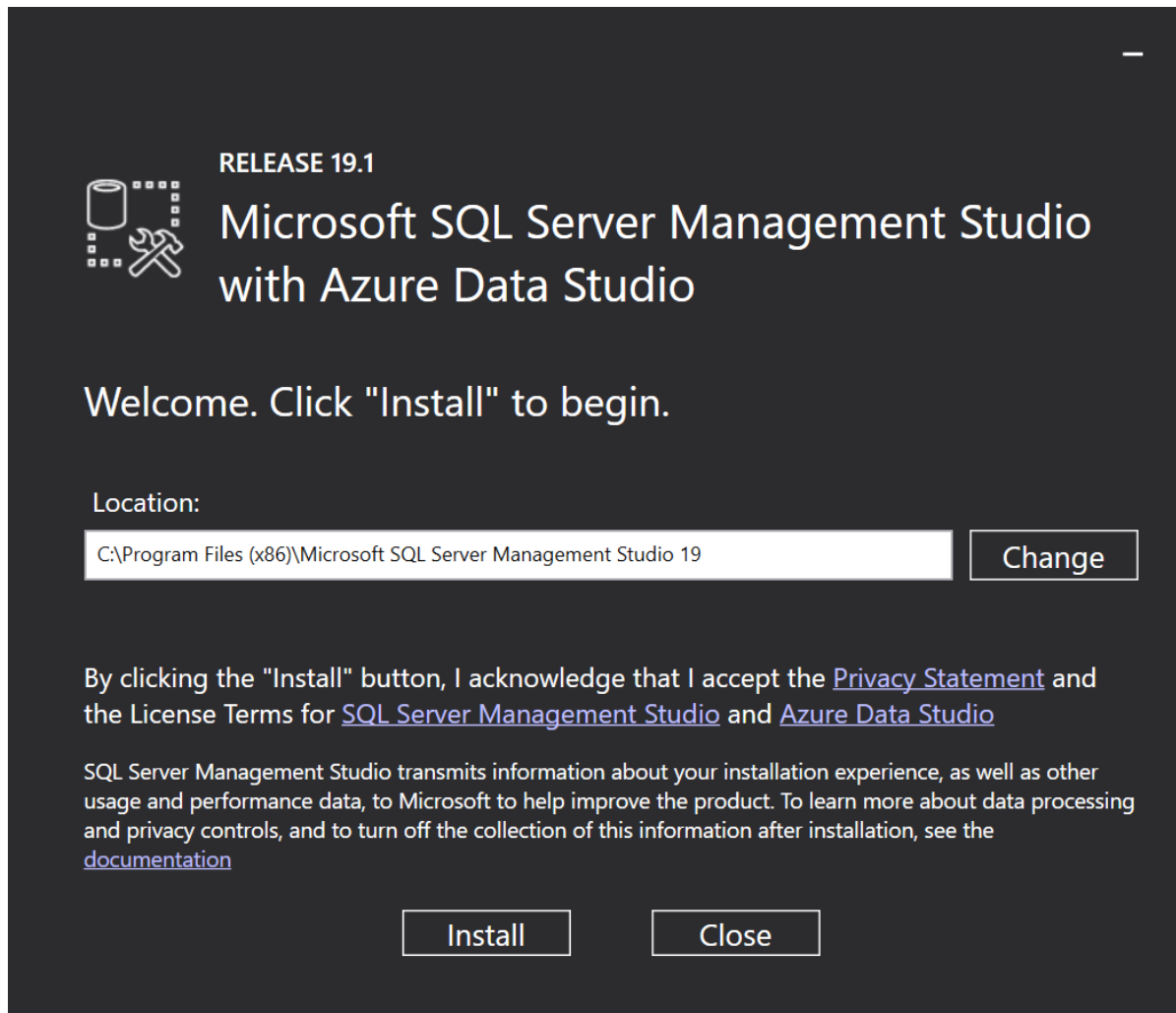
SSMS 19.1 is the latest general availability (GA) version. If you have a *preview* version of SSMS 19 installed, you should uninstall it before installing SSMS 19.1. If you have SSMS 19.x installed, installing SSMS 19.1 upgrades it to 19.1.

- Release number: 19.1
- Build number: 19.1.56.0
- Release date: May 24, 2023

- The latest version is labeled Microsoft SQL Server Management Studio **v19.0.2**.
- Double-click the installation file **SSMS-Setup-ENU.exe** to launch the SSM installer



- Click the Install button



- Wait for a few minutes while the installer sets up the software
- Once setup is completed, click the Close button
- Now, you should have SQL Server 2022 and SQL Server Management Studio installed on your computer.
- Use **SQL Server Configuration Manager** to start or stop SQL Server instance
- OR Start menu > services > MS sqlserver > start/stop service

2.3 Connect to SQL Server

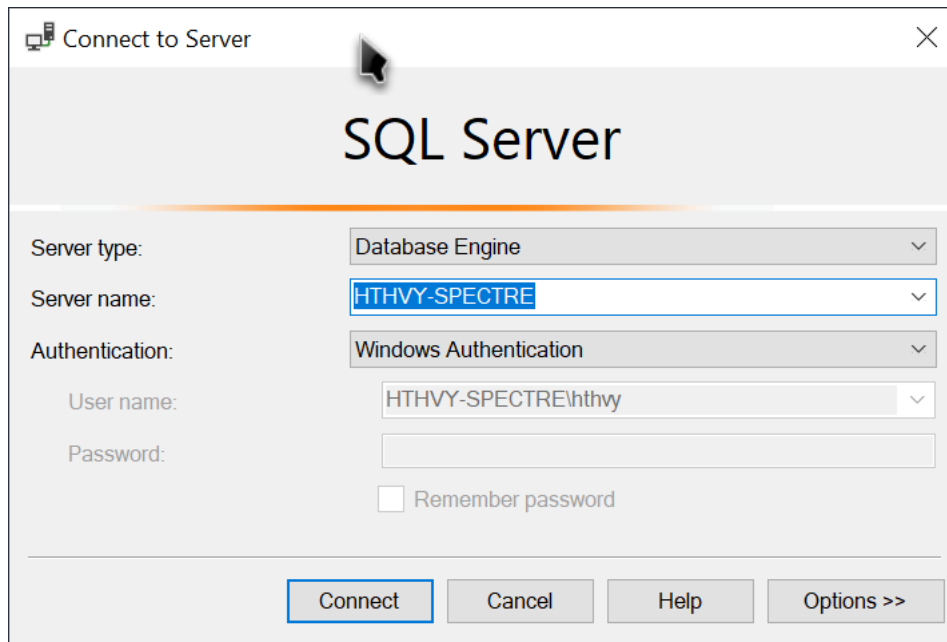
SQL Server authentication ways

There are 2 main authentication ways to connect to an SQL Server instance, you should try:

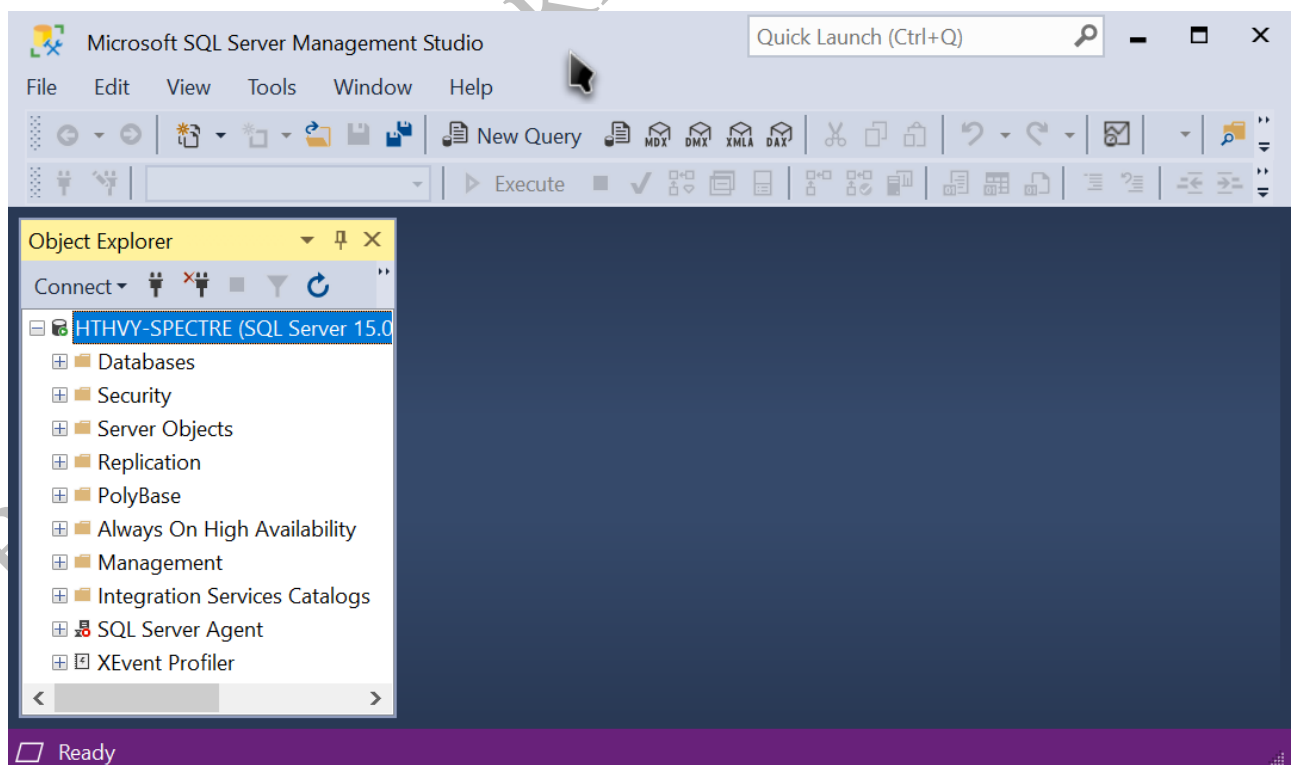
- **Windows Authentication:** Only applicable for SQL Server for Windows, just use the same Windows machine to connect

Username/Password: Applicable for all SQL Server. Default user is **sa** (super admin), password can be set while installing (docker way) or use tools (after login successfully)

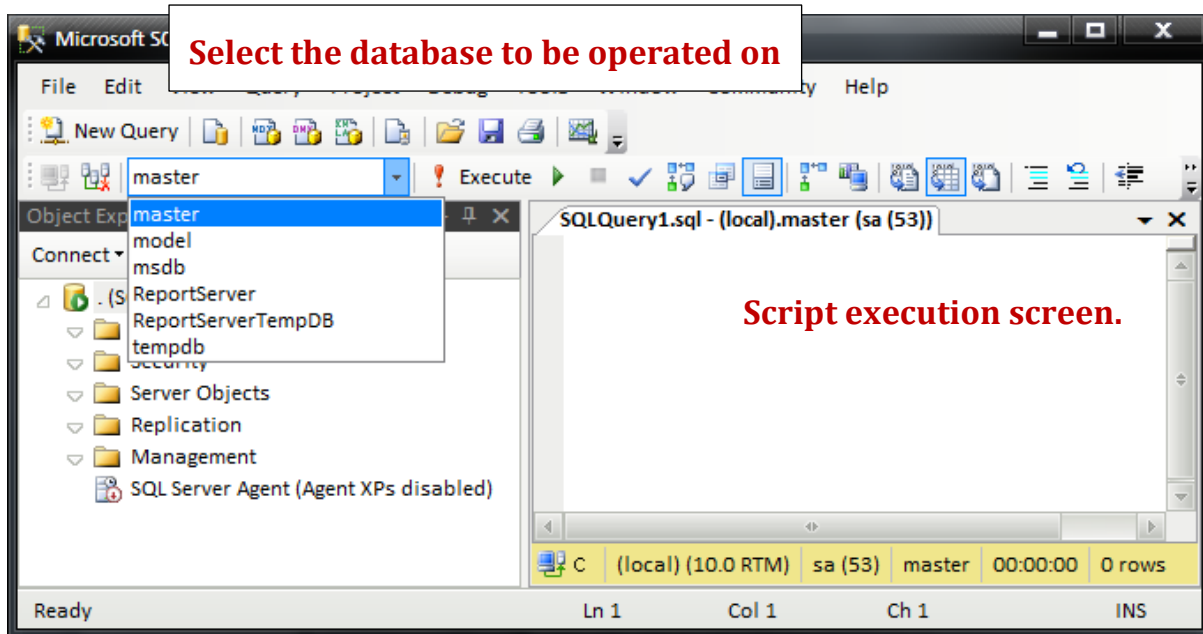
Connect to SQL server using **Windows Authentication**



SQL Management Studio



Interface after connecting to SQL server

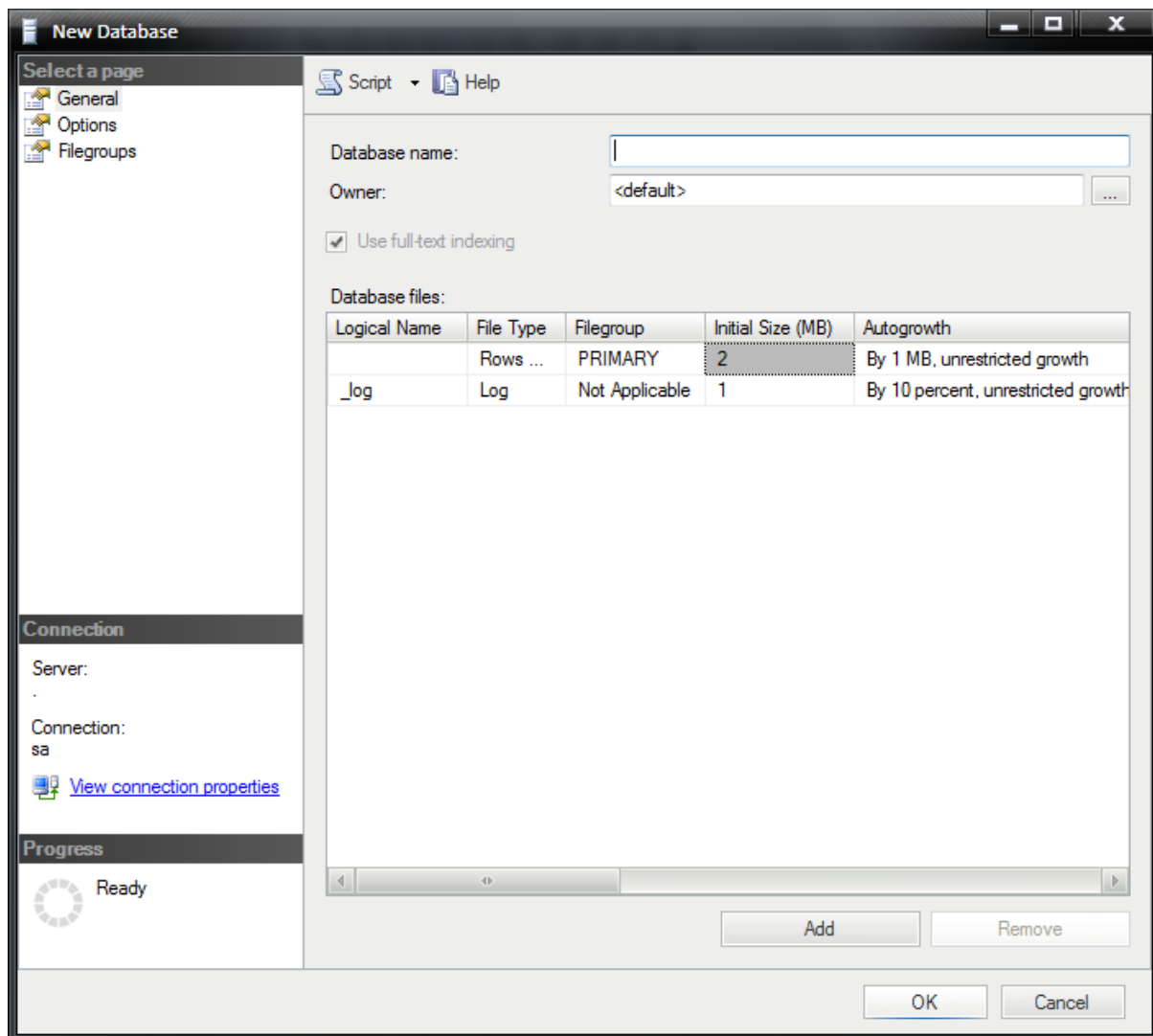


2.4 Basic Operations Using the SQL Server GUI

2.4.1 Creating a New Database:

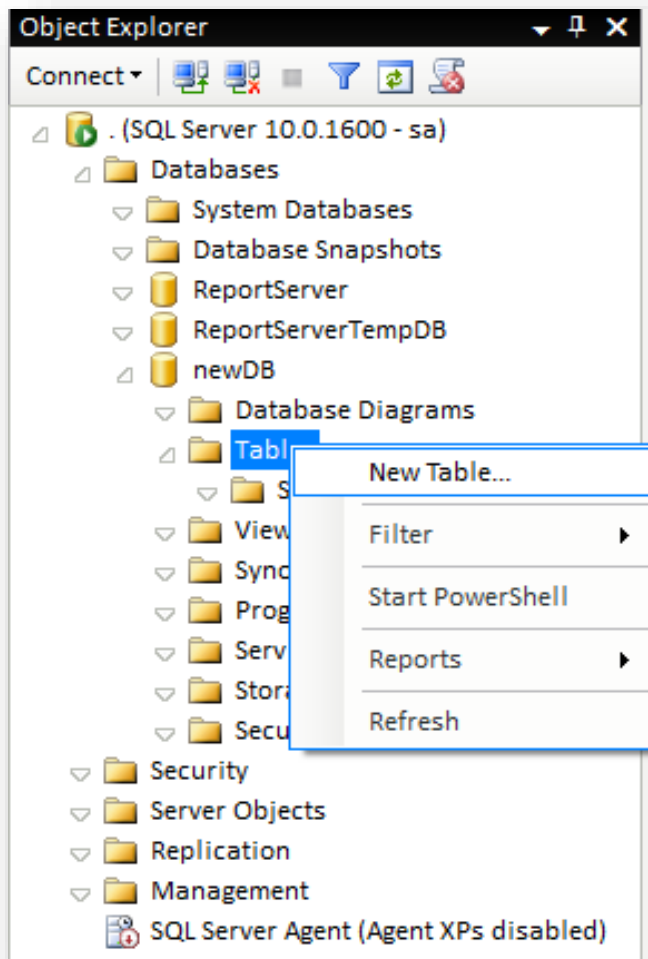
Right-click on **Databases > New Database**. A dialog box appears allowing you to enter a name for the database you want to create. SQL Server will automatically create two files (.mdf: data file and .ldf: log file). You can specify the initial size for the newly

created database; if not specified, SQL Server will use the default settings.



2.4.2 Creating Tables, Constraints, and Inputting Data

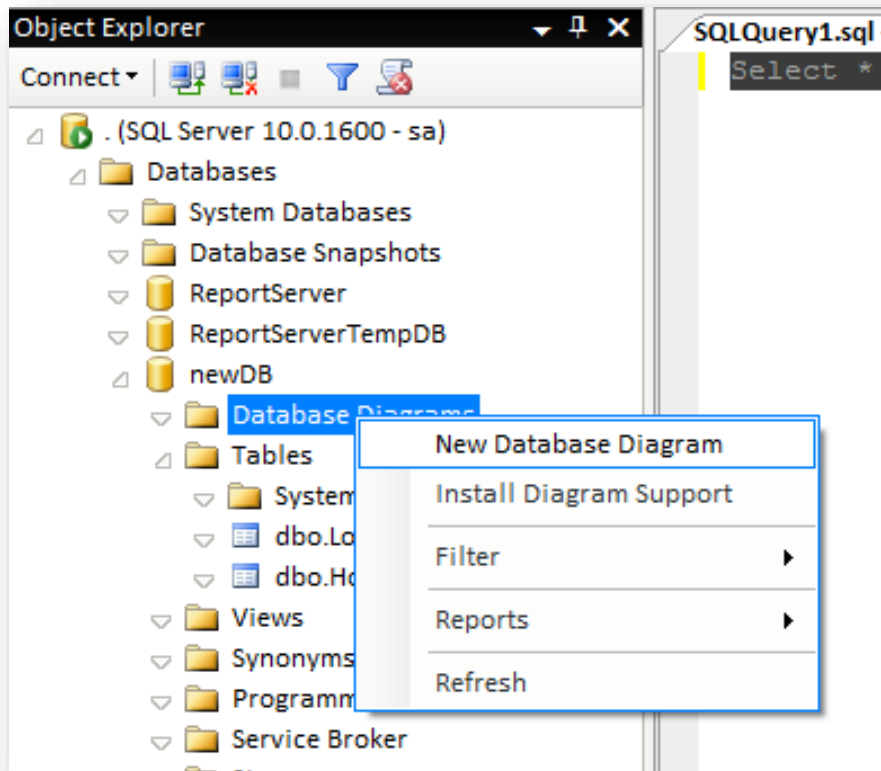
Creating tables



Interface for designing columns, data types, and data constraints (NULL/NOT NULL, UNIQUE, PRIMARY KEY...).

BLUEZERO-PC.newDB - dbo.HocSinh			
	Column Name	Data Type	Allow Nulls
	mahs	int	☐
	tenhs	nchar(10)	☒
	diachi	nchar(10)	☒
	malop	int	☒
			☐

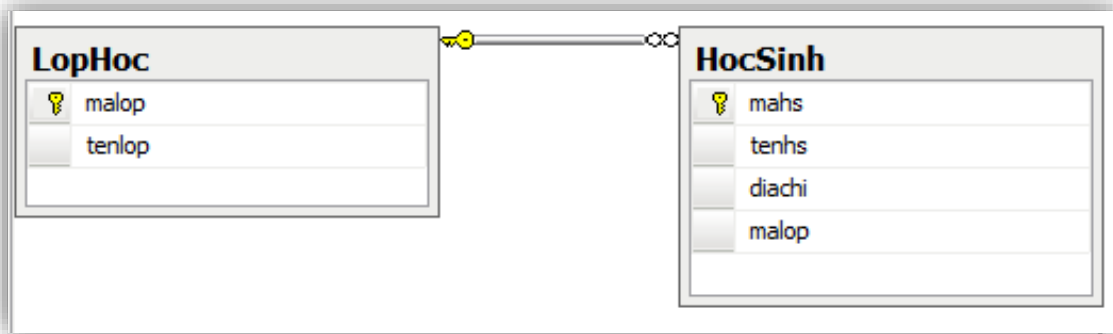
Creating FOREIGN KEY



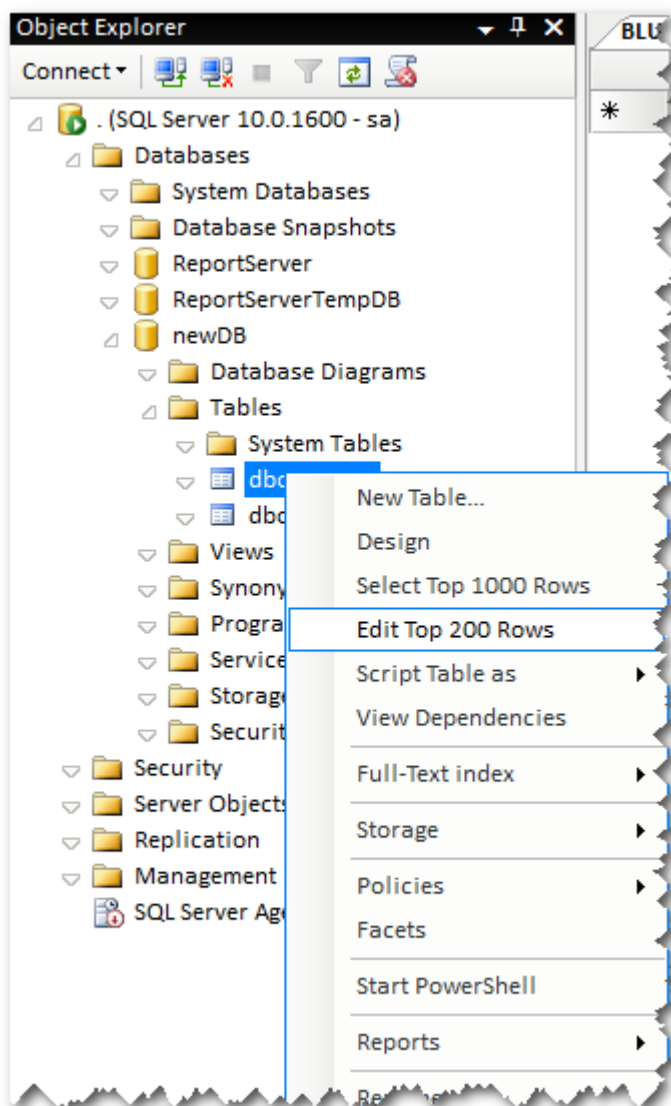
A dialog box appears allowing the selection of tables for which constraints are to be created. The selected tables are then displayed on the screen as follows.



For example, to create a relationship between HocSinh and LopHoc, drag and drop the malop field from the HocSinh table onto the malop field of the LopHoc table. The resulting interface is shown below:

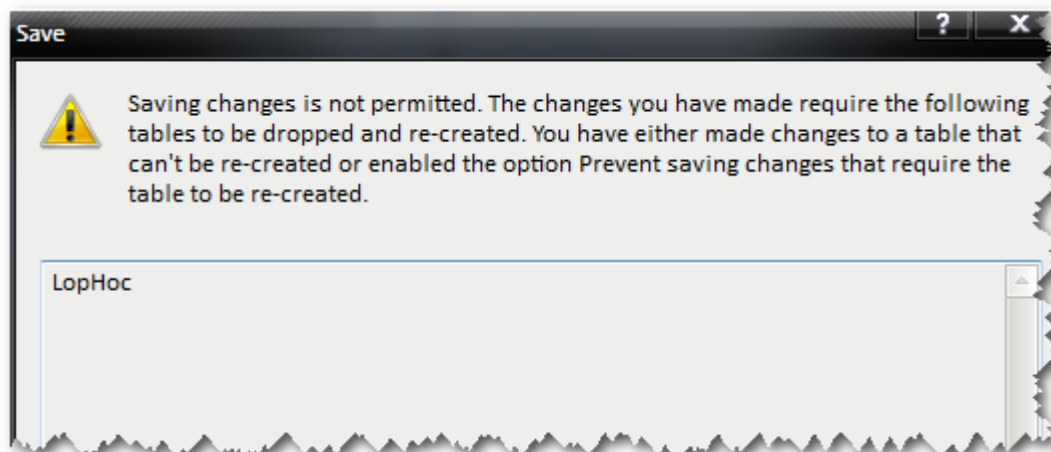


Inputting data:

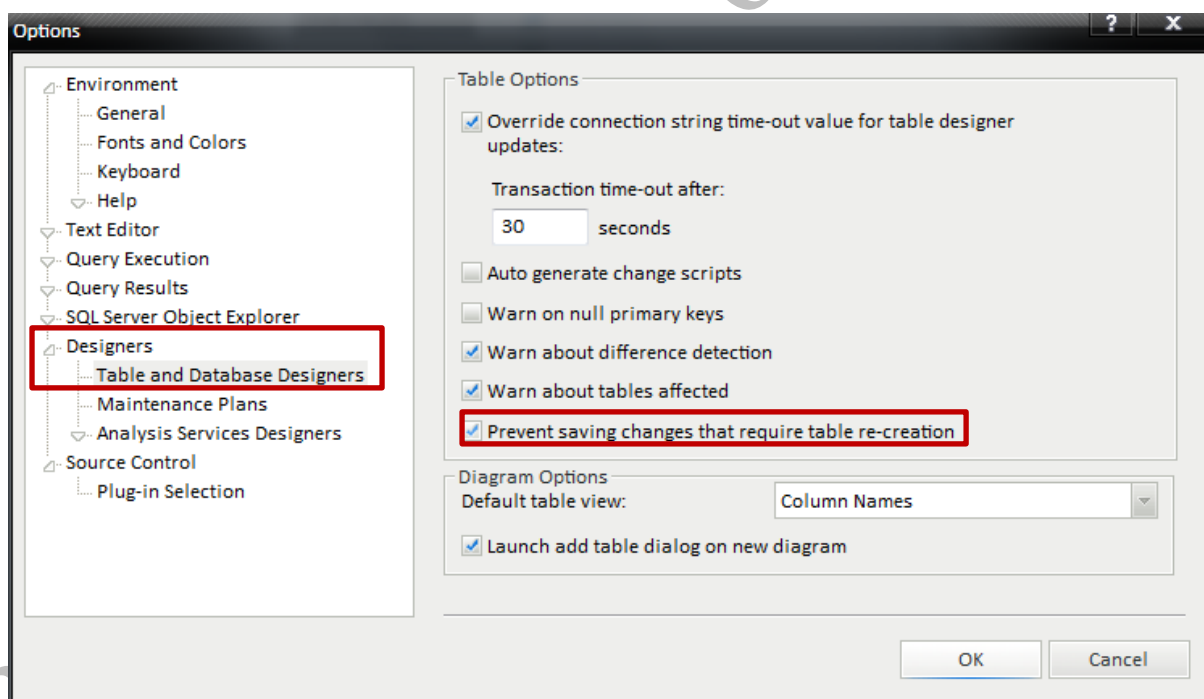


Notes: In some cases, when attempting to modify information (such as column names or data types), an error message may appear and prevent the modification. To make

changes in such situations, the existing object must be deleted and then recreated, as illustrated below.



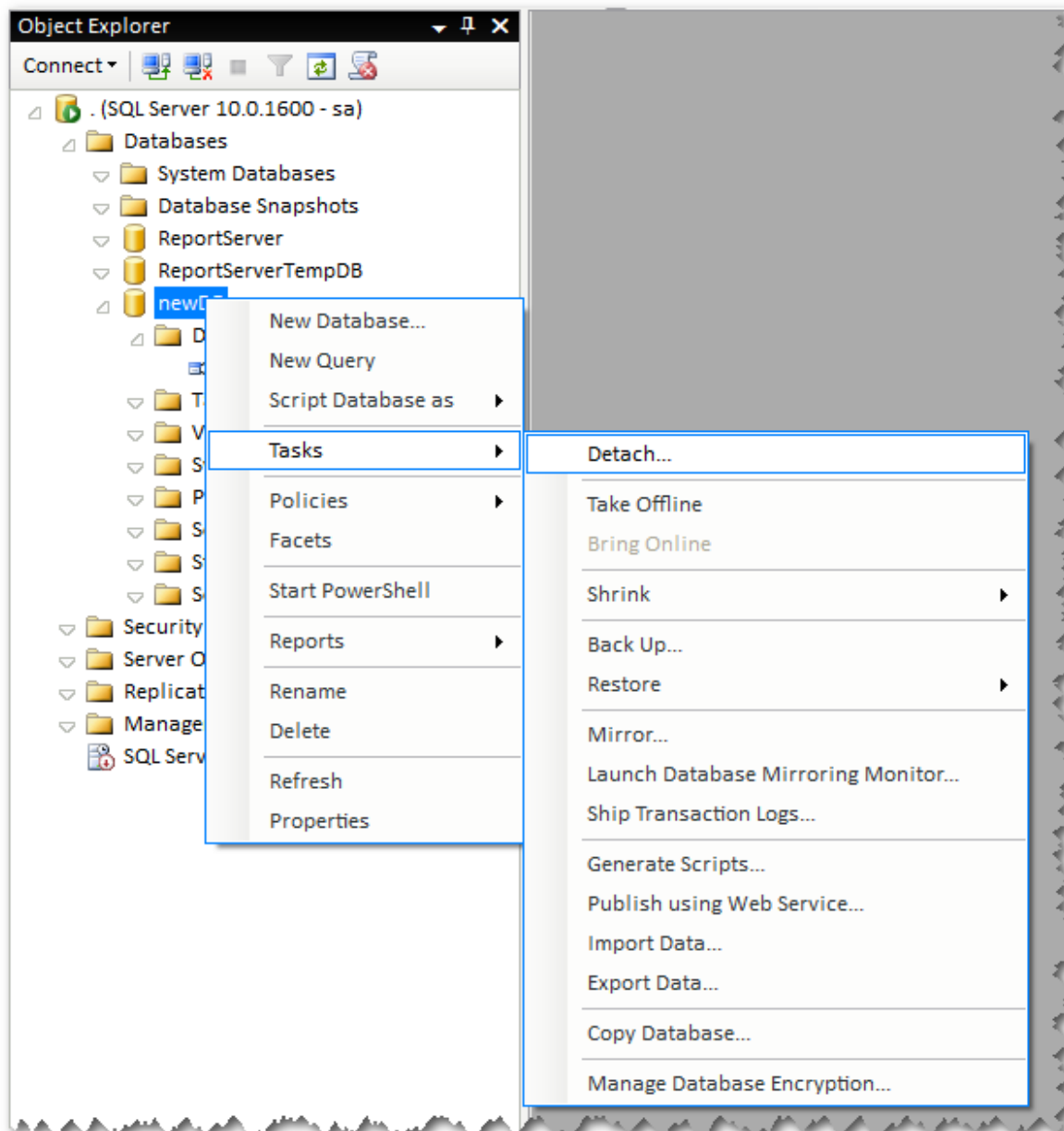
Therefore, to disable this feature, go to the menu Tools > Options > Designer > Table and Database Designers, and uncheck the option "Prevent saving changes that require table re-creation".



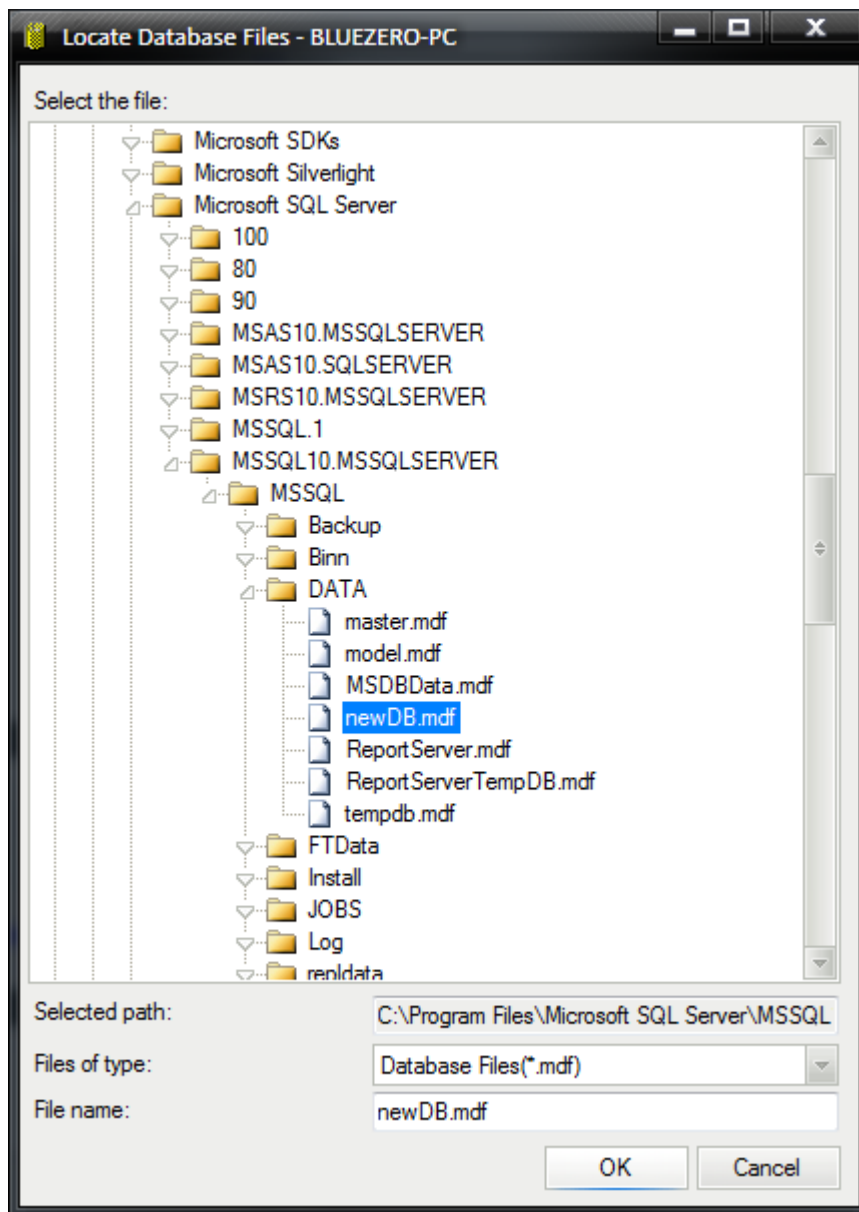
2.4.3 Attach/detach a database

This feature allows a database to be transferred from one machine to another.

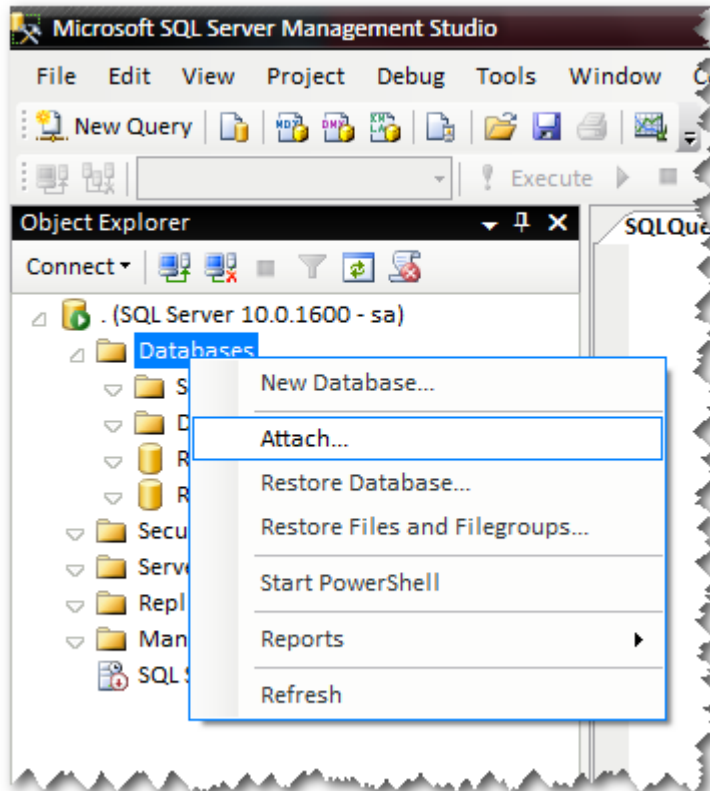
Detach: To perform this operation, ensure that the database to be detached is not the current (active) database. Then proceed as follows:



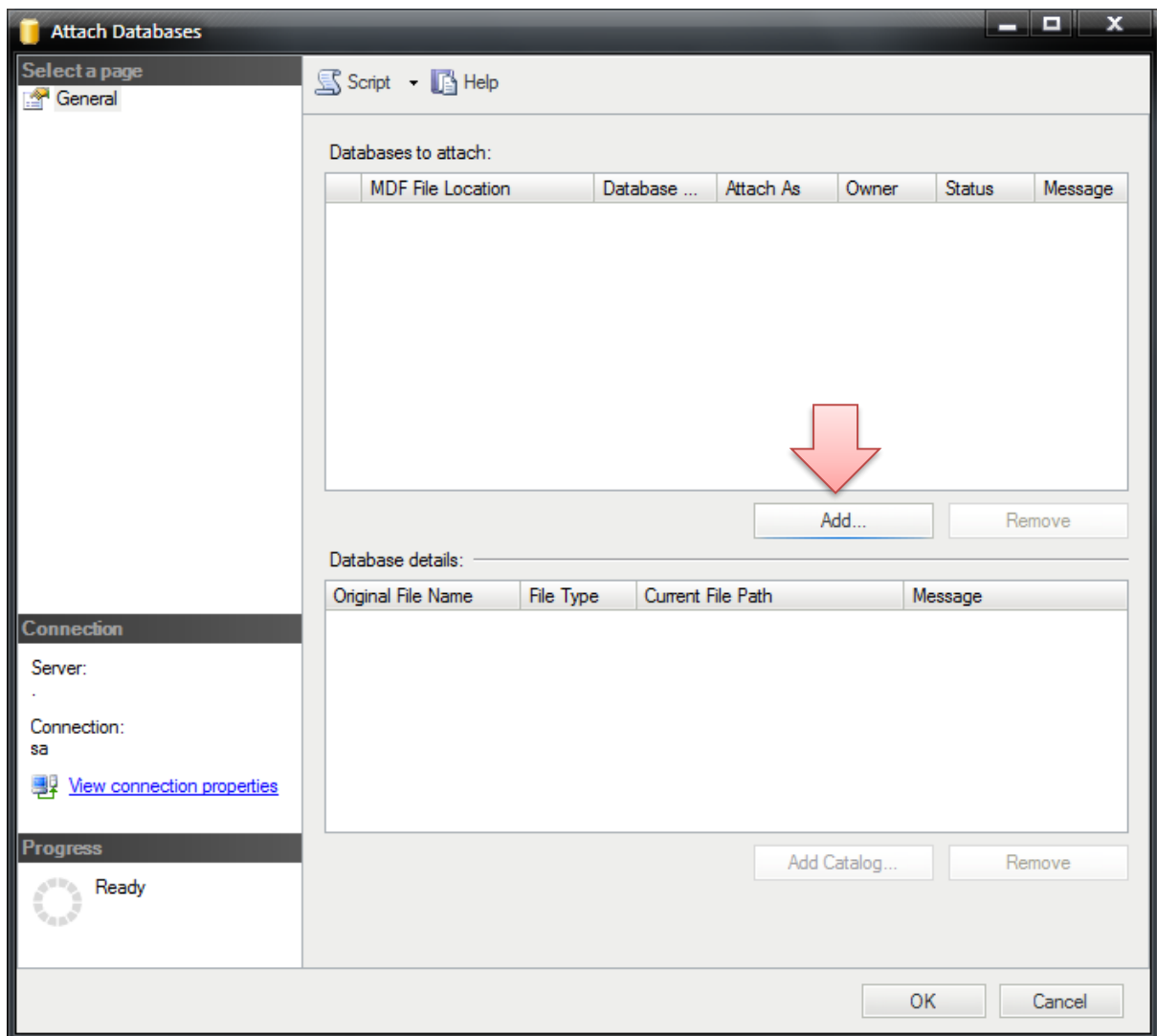
After a successful detach operation, SQL Server will disconnect from the detached database. You can then access the SQL Server installation directory (as shown below) to copy the two files (.mdf and .ldf) and transfer them to another machine, where the database can be recreated by performing an “attach” operation.



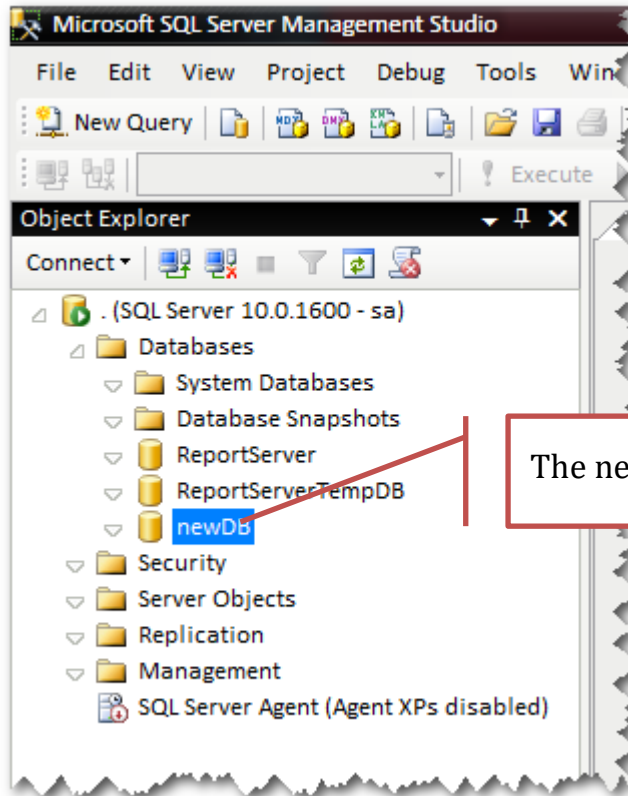
Attach: Right-click on Databases and select Attach



A dialog box appears, prompting you to select the paths to the two files: *.mdf* and *.ldf*.

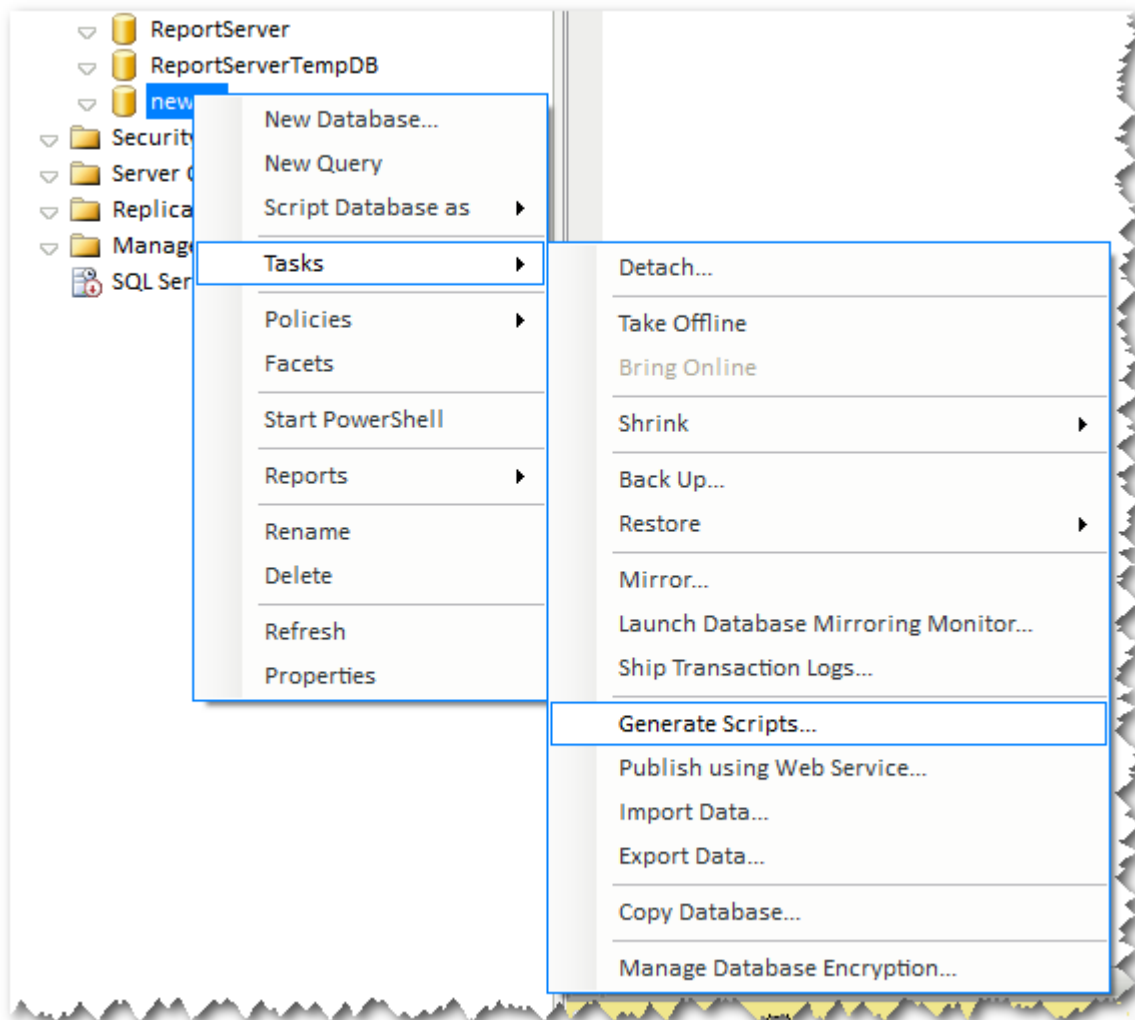


After completion, the newly attached database will appear in the list of databases.

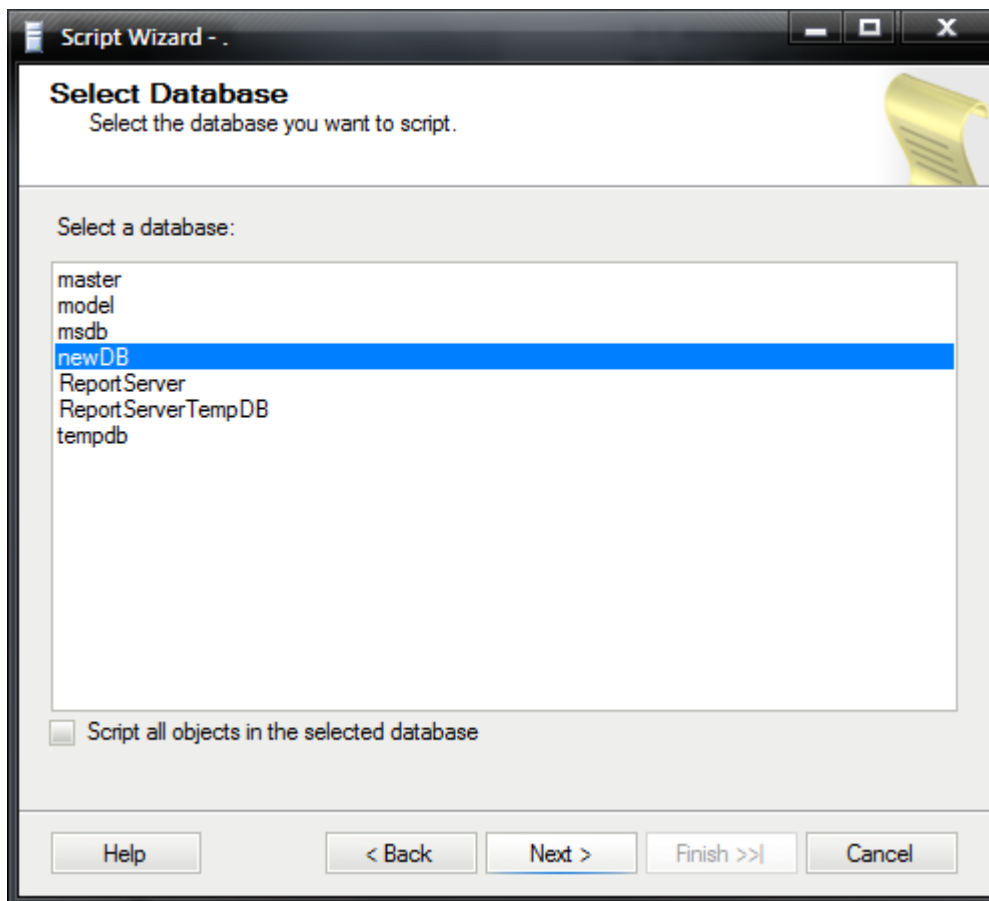


2.4.4 Generating database creation scripts

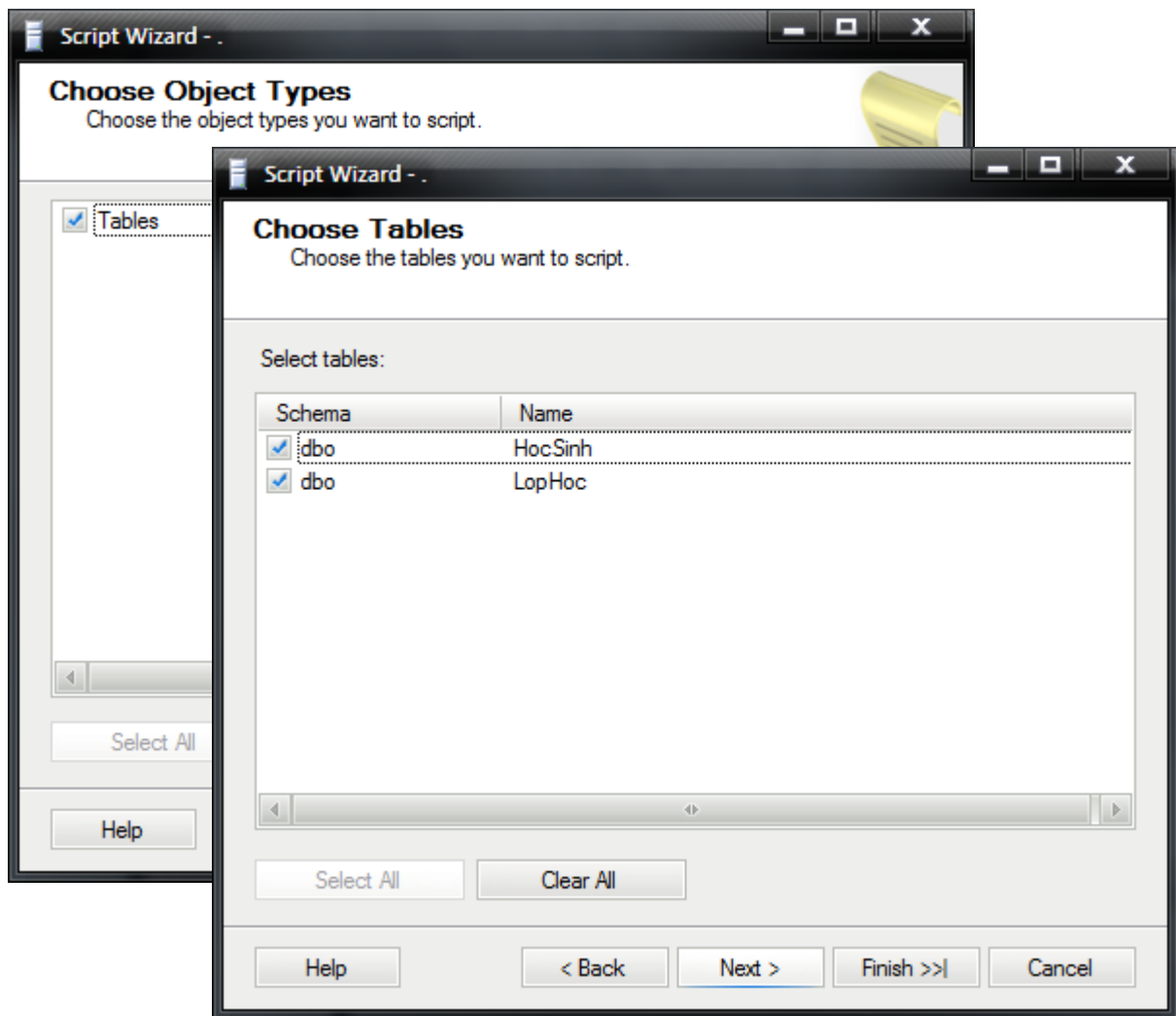
In cases where you want SQL Server to generate scripts for creating a database (without generating data insertion scripts), use the **“Generate Scripts”** feature as follows:



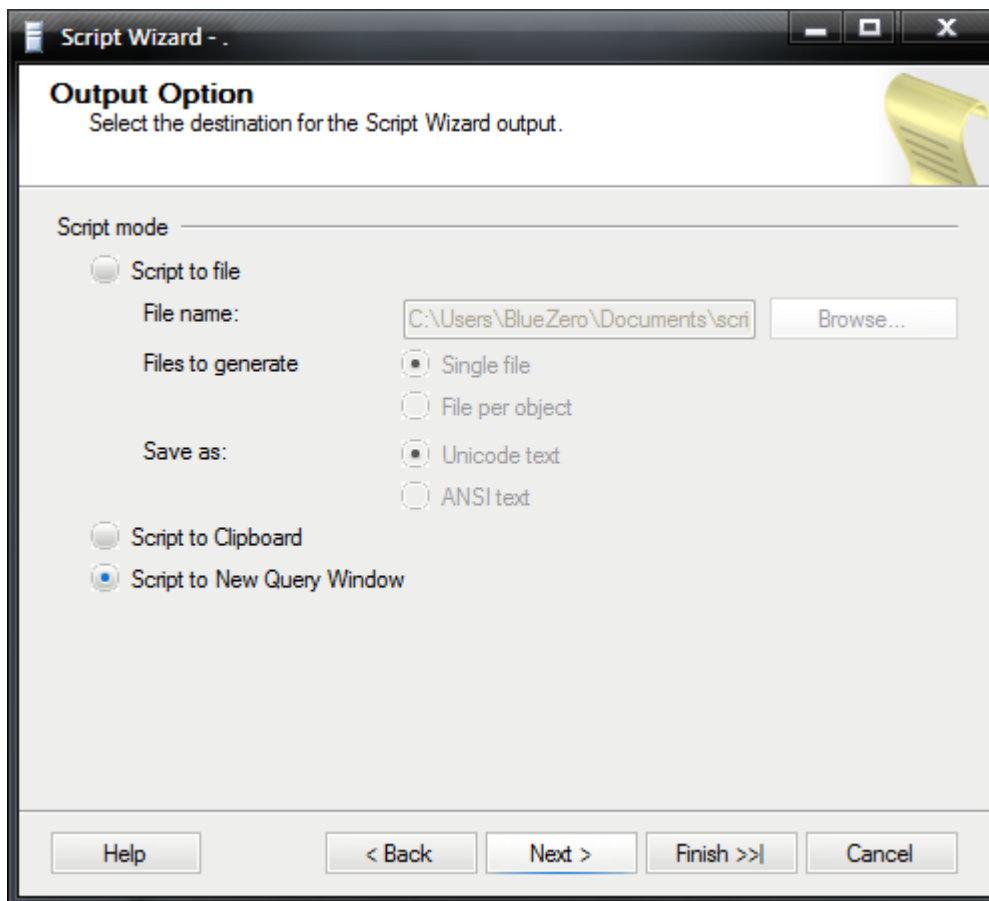
A dialog box appears allowing the generation of scripts. Click **Next**. Select the database for which the script is to be generated.



After choosing the database and clicking Next, a screen is displayed that allows you to select the tables, stored procedures, and functions (if any have been created previously) for which scripts will be generated.



Select the script generation option: save the script to a file (either as a single file or multiple files), save it to memory, or open it directly in the SQL Server script execution window.



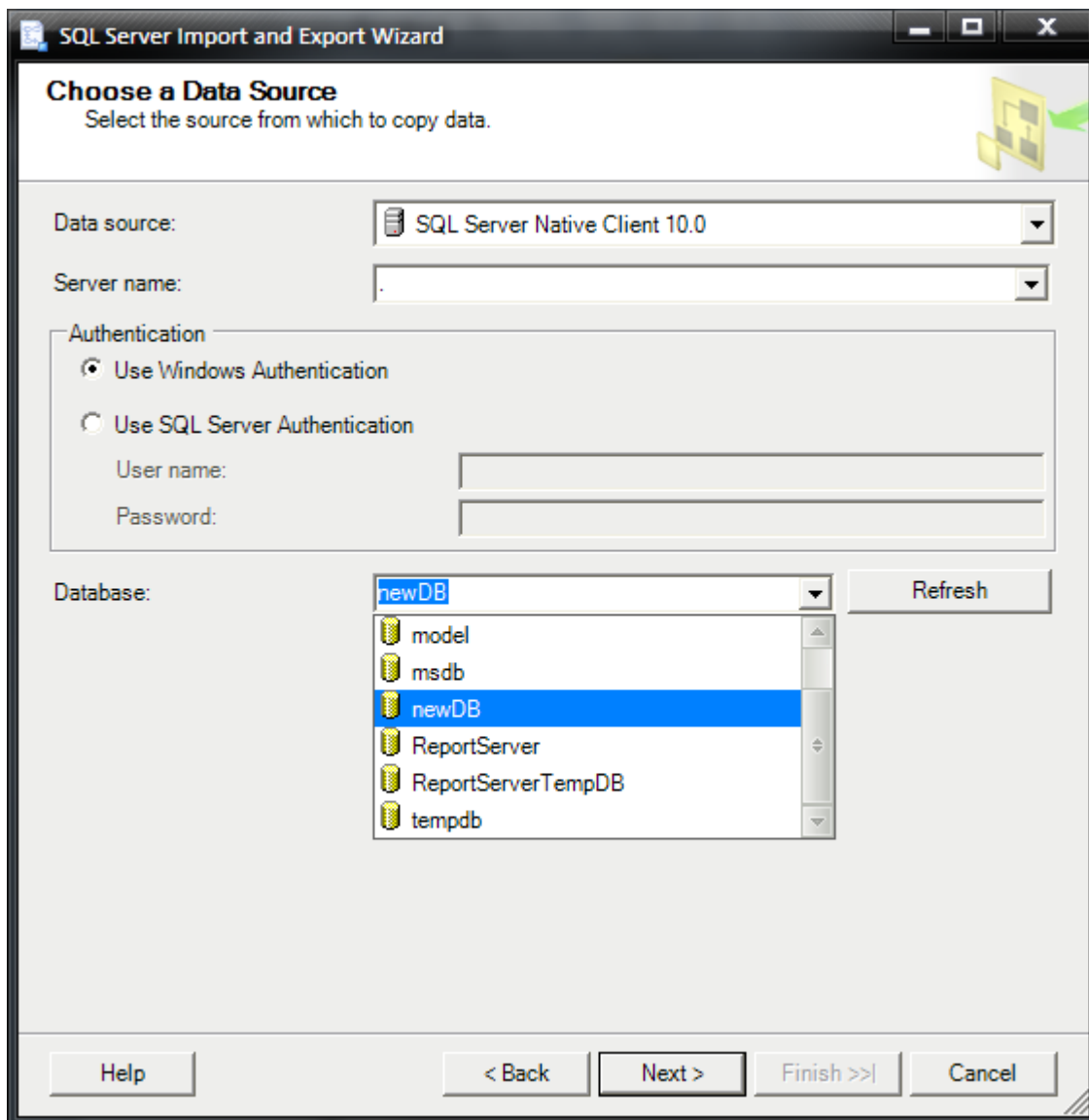
Result:

```
SQLQuery3.sql - (local).master (sa (58)) SQLQuery2.sql - (local).master (sa (53))
USE [newDB]
GO
/***** Object: Table [dbo].[LopHoc]      Script Date: 01/27/2013 19:42:28 *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[LopHoc] (
    [malop] [int] NOT NULL,
    [tenlop] [nvarchar](50) NULL,
    CONSTRAINT [PK_LopHoc] PRIMARY KEY CLUSTERED
    (
        [malop] ASC
    )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, AL
) ON [PRIMARY]
GO
/***** Object: Table [dbo].[HocSinh]      Script Date: 01/27/2013 19:42:28 *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[HocSinh] (
    [mahs] [int] NOT NULL,
    [tenhs] [nchar](10) NULL,
    [diachi] [nchar](10) NULL,
    [malop] [int] NULL,
    CONSTRAINT [PK_HocSinh] PRIMARY KEY CLUSTERED
    (
        [mahs] ASC
    )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, AL
) ON [PRIMARY]
GO
```

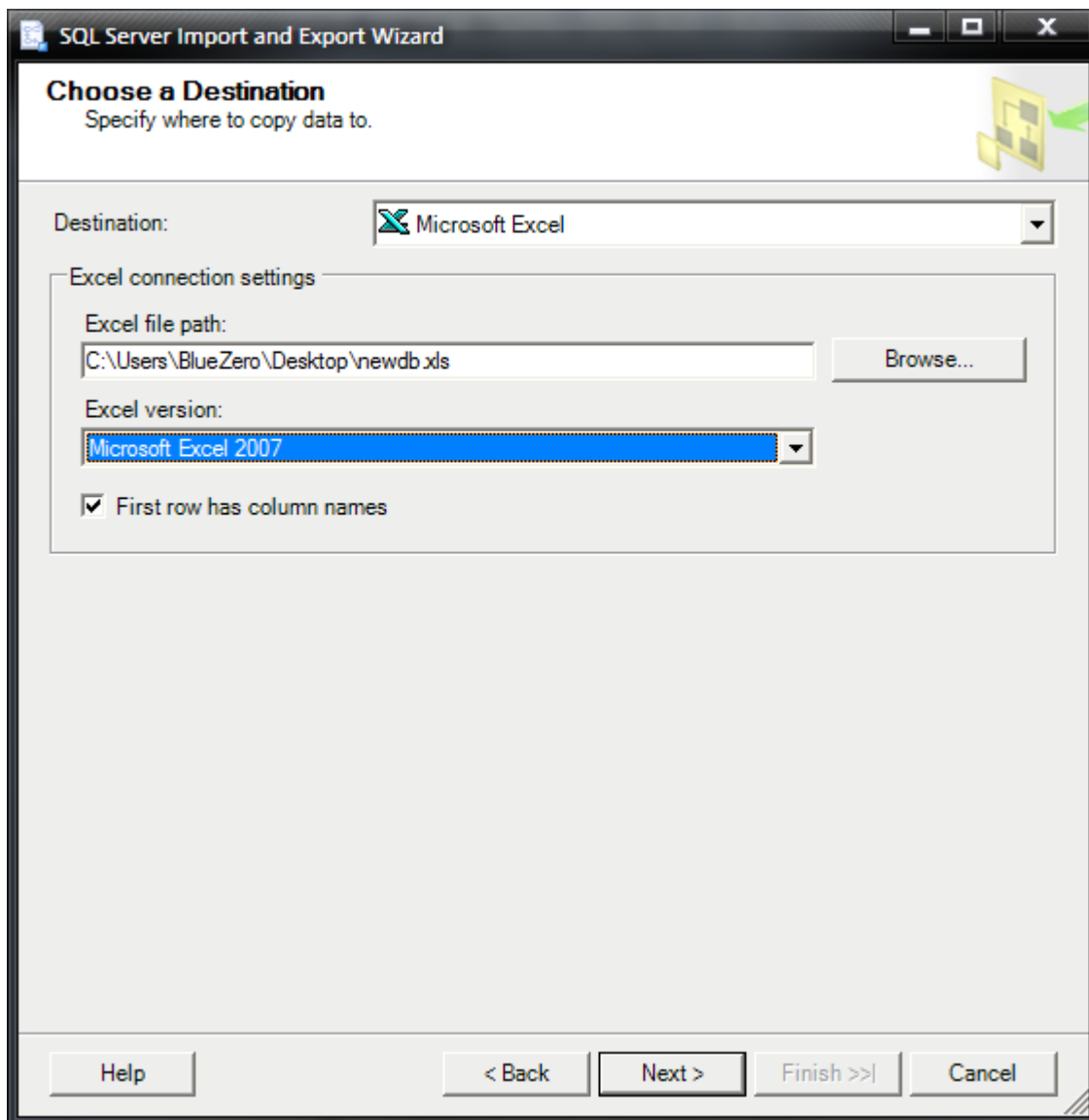
2.4.5 Data Export/Import

This feature allows data to be exported to various formats (such as text files, Microsoft Access, Excel, SQL Server, etc.).

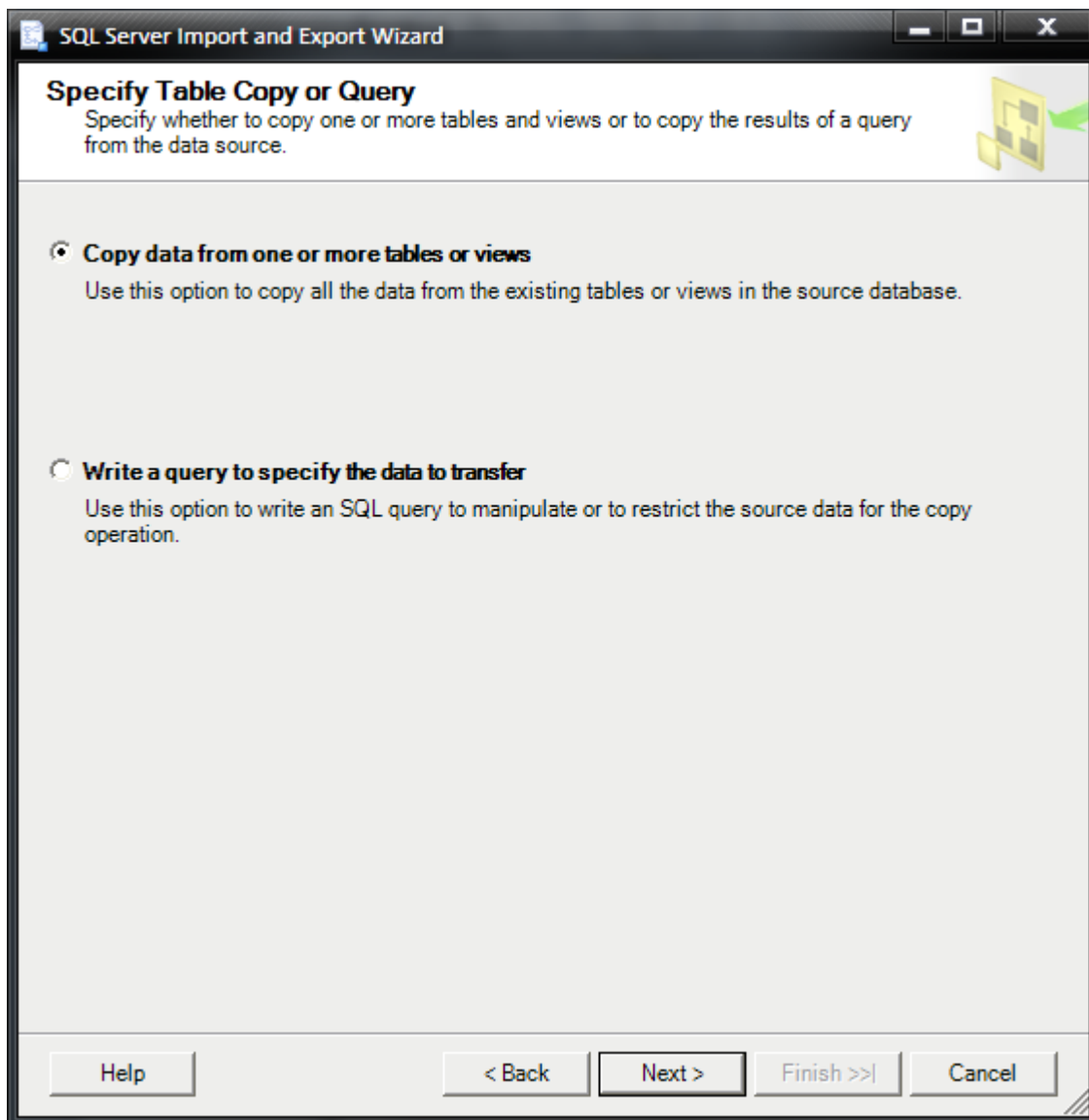
Export: Right-click on the database > **Tasks > Export Data**. Enter the connection information and select the database to be exported.



Select and specify the type and storage location of the destination database.



Click **Next**, then choose the data transfer method: copying data from tables/views in the source database or writing a query to retrieve data from the source. In this case, the first method is selected.



Then click **Next** and select the tables or views to be exported. Below is the result of the data exported to an Excel file, which includes two sheets corresponding to the two tables selected for export.

	A	B	C	D
1	mahs	tenhs	diachi	malop
2		1 Nguyen A	DBP	1
3		2 Tran B	NTMK	2

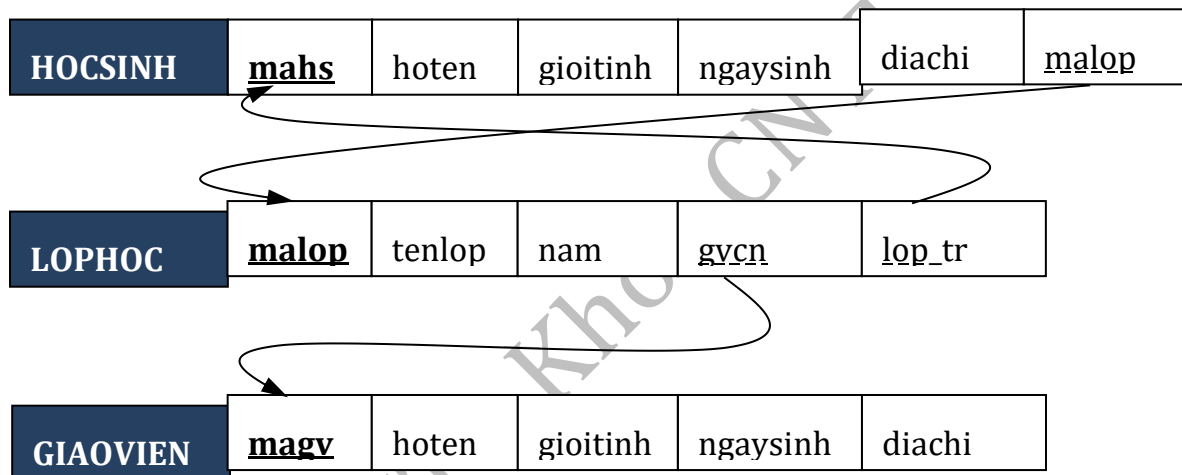
Excel sheets: HocSinh, LopHoc

Import: Performed in a similar manner.

3 In-class exercise

Requirements:

- Study how to connect to and use SQL Server in the computer lab.
- Study how to start and stop services.
- Open and review the online documentation (online book).
- Use the graphical interface to create a database named QLLOpHoc with the following table structure:



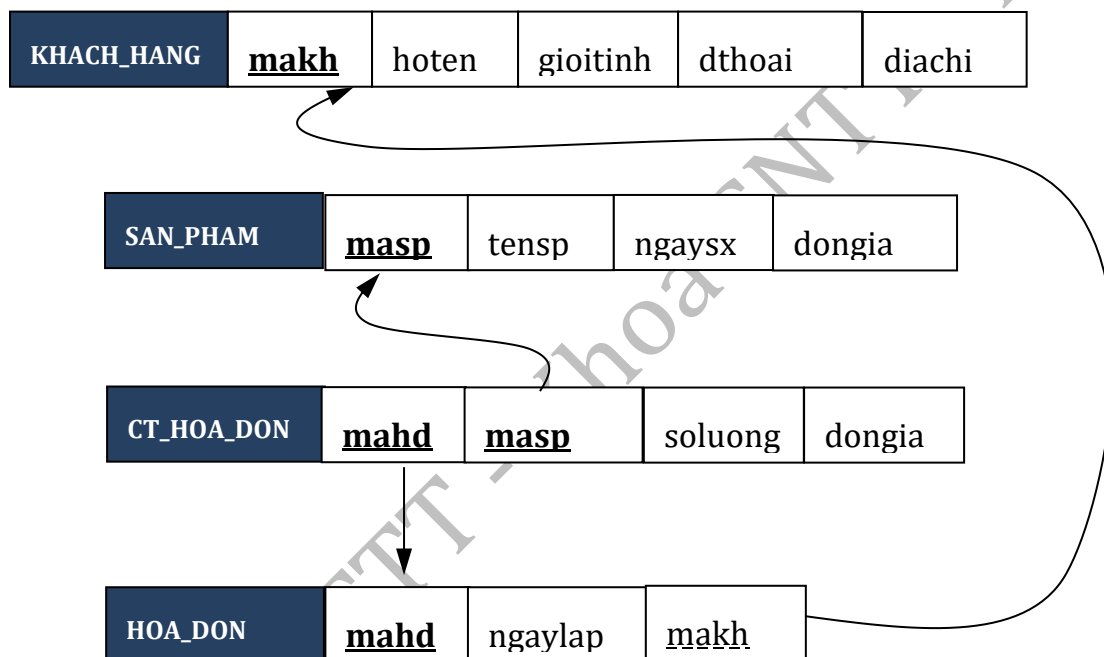
- Enter data into the QLLOpHoc database created above (each table must contain at least 5 rows of data).
- Perform detach / attach operations on the QLLOpHoc database.
- Generate scripts for QLLOpHoc into two files: CauTruc.sql (storing scripts for creating the database structure and constraints) and DuLieu.sql (storing scripts for inserting data into the QLLOpHoc database).
- Export the data of QLLOpHoc to an Excel file.

Duration: 02 hours.

4 Homework exercise

Requirements:

- Download and install SQL Server.
- Study the GUI components in SQL Server.
- Use the graphical interface to create a database named *QLBanHang* with the following table structure:



- Enter data into the *QLBanHang* database created above (each table must contain at least 5 rows of data).
- Perform detach / attach operations on the *QLBanHang* database.
- Generate scripts for *QLBanHang* into two files: *CauTruc.sql* (containing scripts for creating the database structure and constraints) and *DuLieu.sql* (containing scripts for inserting data into the *QLBanHang* database).

- g. Export the data of QLBanHang to an Excel file.

Duration: 03 hours.

Bộ môn HTTT - Khoa CNTT - ĐH KHTN